

具身智能

第2讲：具身智能系统概览

Junwei Liang
AI Thrust
香港科技大学（广州）

Embodied AI

Lecture 2: Embodied AI System Overview

Junwei Liang
AI Thrust
HKUST (Guangzhou)

目录

- 具身智能范式
- 具身智能系统设计
- 传感器概览
- 机器人视觉 vs. 计算机视觉
- 实验室机器人演示

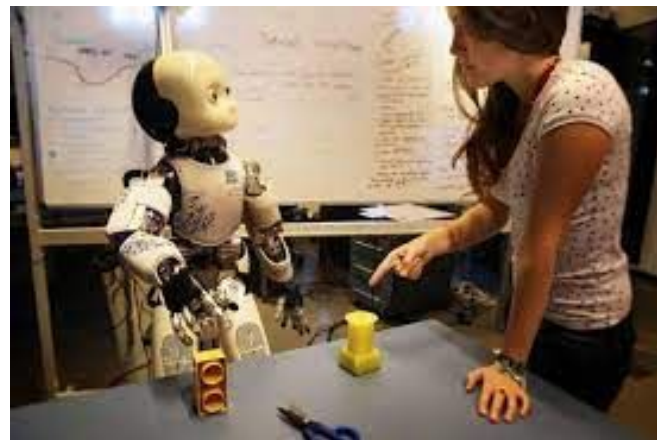
Agenda

- The Embodied AI paradigm
- Embodied AI System Design
- Sensors Overview
- Robot Vision vs. Computer Vision
- Robot Demos from my lab

什么是具身智能？

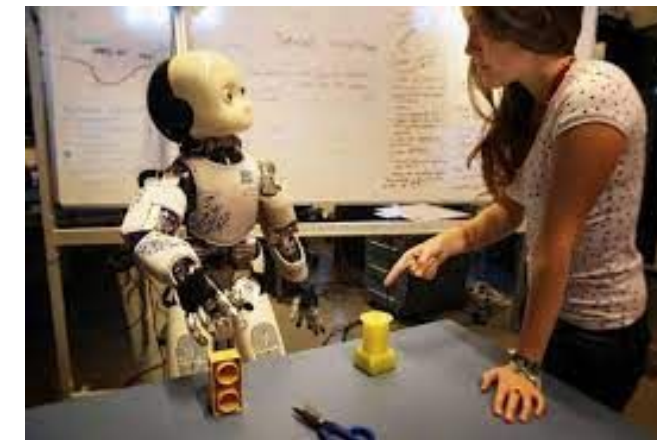
- 具身智能是机器学习与机器人学的交叉
- 研究如何利用传感数据训练最优方案，使机器人行动并完成目标
- 感知与决策是关键
- 可能的未来

○ 通用家用机器人！



What is Embodied AI?

- Embodied AI is the intersection of **machine learning and robotics**
- The study of **training optimal solutions using sensory data** for robots to act and achieve goals
- Perception and decision making are the key
- Possible future
 - General-purpose home robots!



具身智能范式

- 先从一个问题开始——如何打造自动驾驶汽车？

- 已知条件

- 起点、目标位置
- 城市地图
- 传感器：相机、LiDAR 等

- 目标

- 最小化到目标的距离

- 约束

- 沿道路行驶
- 遵守交通规则
- 不与其他物体碰撞



The Embodied AI Paradigm

- Let's start with - How do we make a self-driving vehicle?

- Given

- Starting location, Goal location
- Map of the city
- Sensors - Camera, LIDAR, etc.

- Objective

- Minimize distance to goal

- Constraint

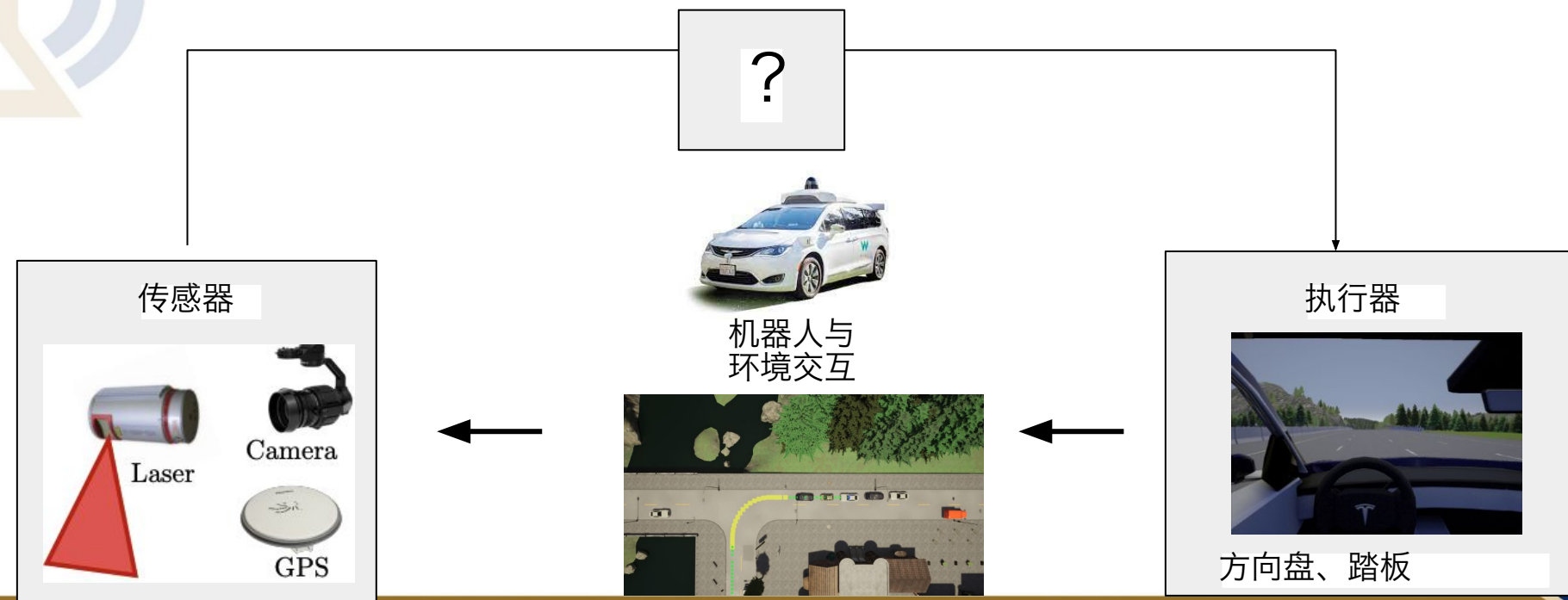
- Follow the road
- Follow traffic rules
- No crashing into other objects



具身智能范式

- 如何打造自动驾驶汽车？——流程图如下

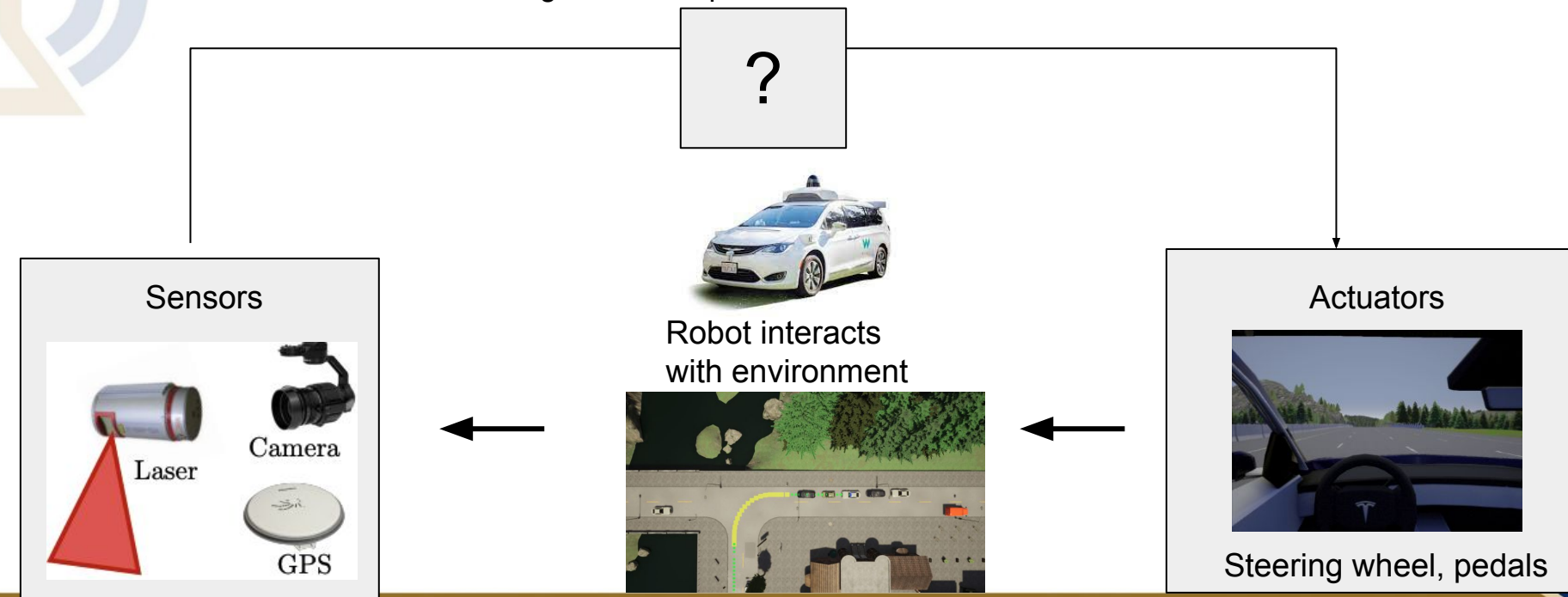
○ 动作包括方向盘和踏板



The Embodied AI Paradigm

- How do we make a self-driving vehicle? - here is a flow chart

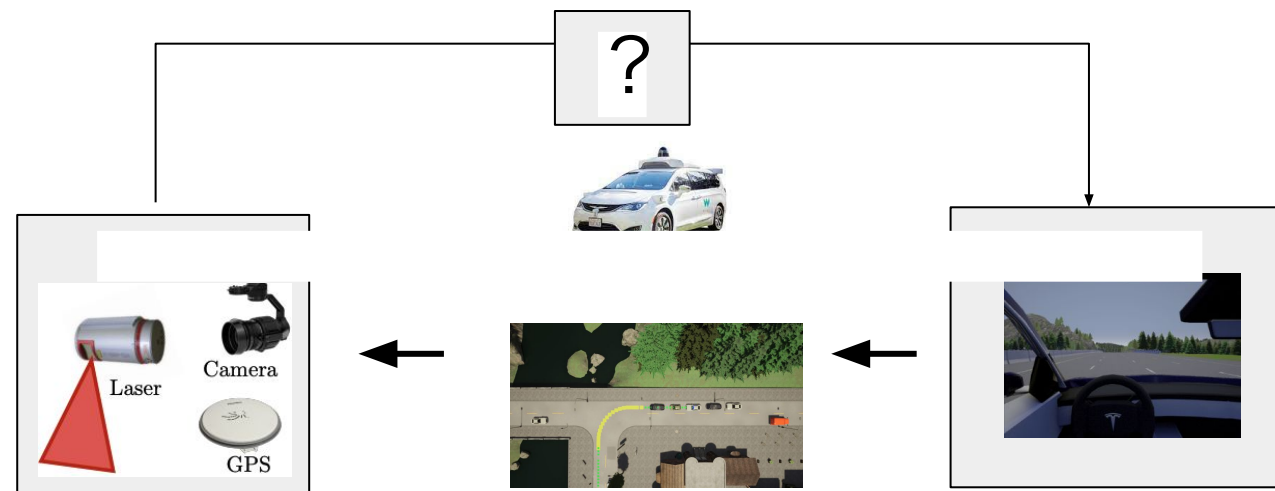
○ Actions includes steering wheel and pedals



具身智能范式

- 关键问题

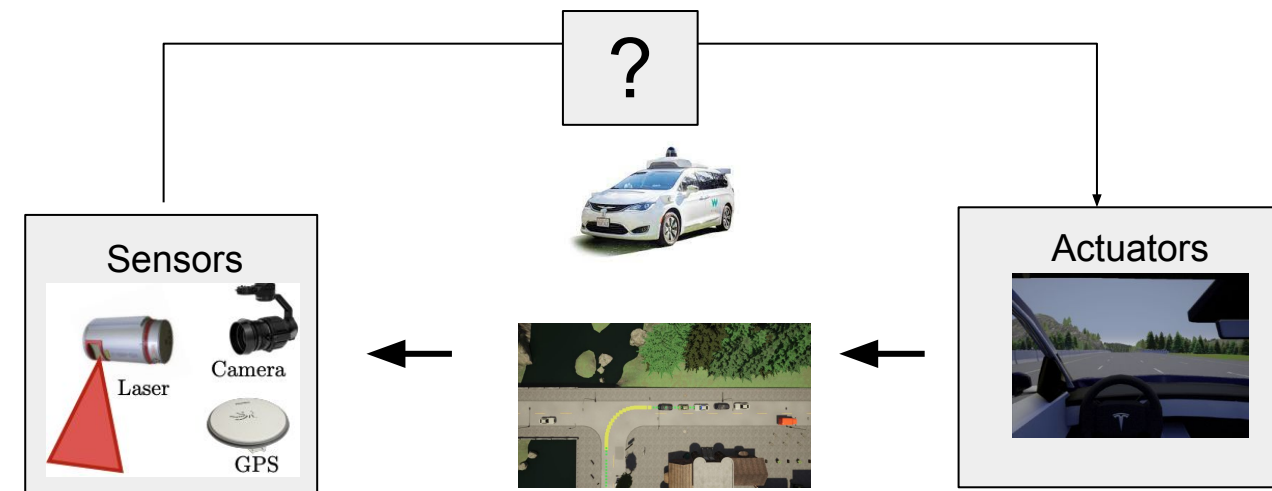
- Q1: 假设我们已知世界的一切。应向执行器发送什么指令?
- Q2: 如何根据传感器更新世界知识?



The Embodied AI Paradigm

- Key questions

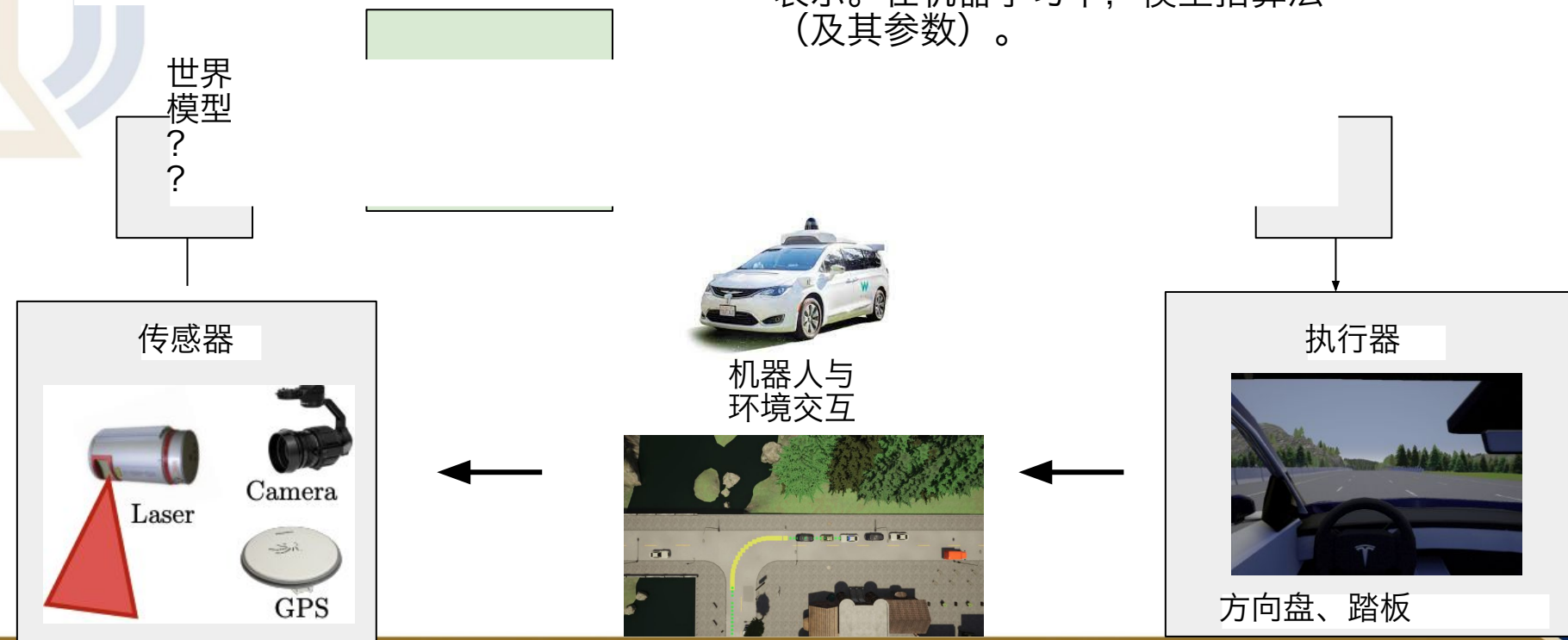
- Q1: **Assume we know everything about the world.** What commands should we send to the actuator?
- Q2: How do we update the world knowledge given the sensors?



具身智能范式

- 假设我们有一个世界模型

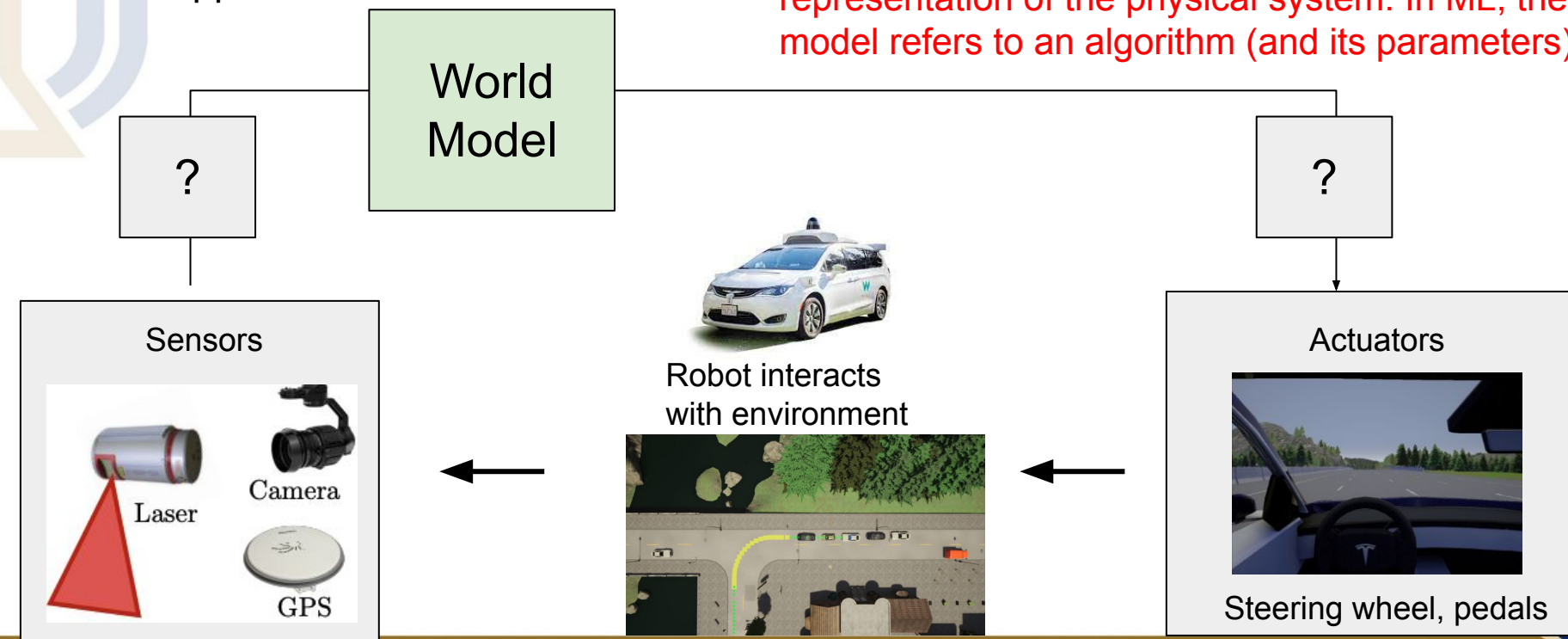
注意：在机器人学中，“模型”指物理系统的表示。在机器学习中，模型指算法（及其参数）。



The Embodied AI Paradigm

- Suppose we have a **world model**

Note in robotics, people use “model” to refer to a representation of the physical system. In ML, the model refers to an algorithm (and its parameters).



具身智能范式

注意：在机器人学中，“模型”指物理系统的表示。在机器学习中，模型指算法（及其参数）。

• 世界模型是什么？它包括：

- 机器人/车辆在哪里？当前状态如何？
- 哪些是障碍（人行道、草地、树木），哪些可以行驶（道路）？
- 需要避开哪些物体（其他车辆、行人）？
- 车辆本身有哪些属性？（车辆模型 / Car Model）

■ 能开多快？加速度多大？

■ 地面是否湿滑？

○ 如今，世界模型（或世界动作模型 WAM）指“像素与动作预测”模型。因此视觉生成模型也被视为世界模型

更精确地说，世界模型是“状态转移”模型 $P(s_t, a_t) = s_{t+1}$ 它预测未来的“状态”。

“WAM”指世界动作模型 $P(s_t, a_t) = (s_{t+1}, a_t)$ ，同时预测未来状态与动作

The Embodied AI Paradigm

Note in robotics, people use “model” to refer to a representation of the physical system. In ML, the model refers to an algorithm (and its parameters).

• What is the world model? It includes:

- **Where is the robot/vehicle? What is its current status?**
- What are obstacles (sidewalk, grass, trees) and what are available to drive (road)?
- What objects to avoid (other vehicle, pedestrians)?
- What are the properties of the vehicle? (Car Model)
 - How fast it can go? Acceleration rate?
 - Wet ground?
- Nowadays, **world model (or world action model, WAM)** means “pixel and action prediction” model. So the vision generative models are considered world model

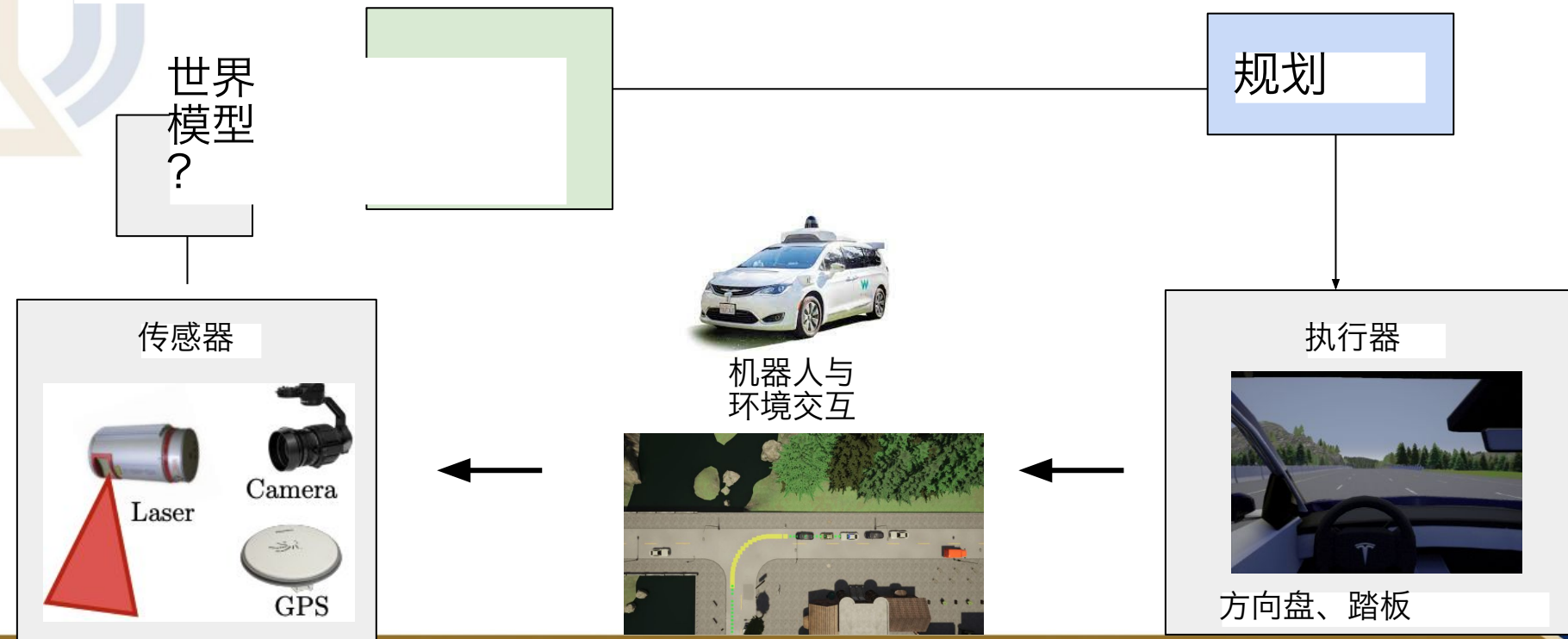
To be more precise, world model means a “state transition” model $P(s_t, a_t) = s_{t+1}$

It predicts the future “state”.

“WAM” refers to world action model $P(s_t, a_t) = (s_{t+1}, a_t)$, it predicts both future states and actions

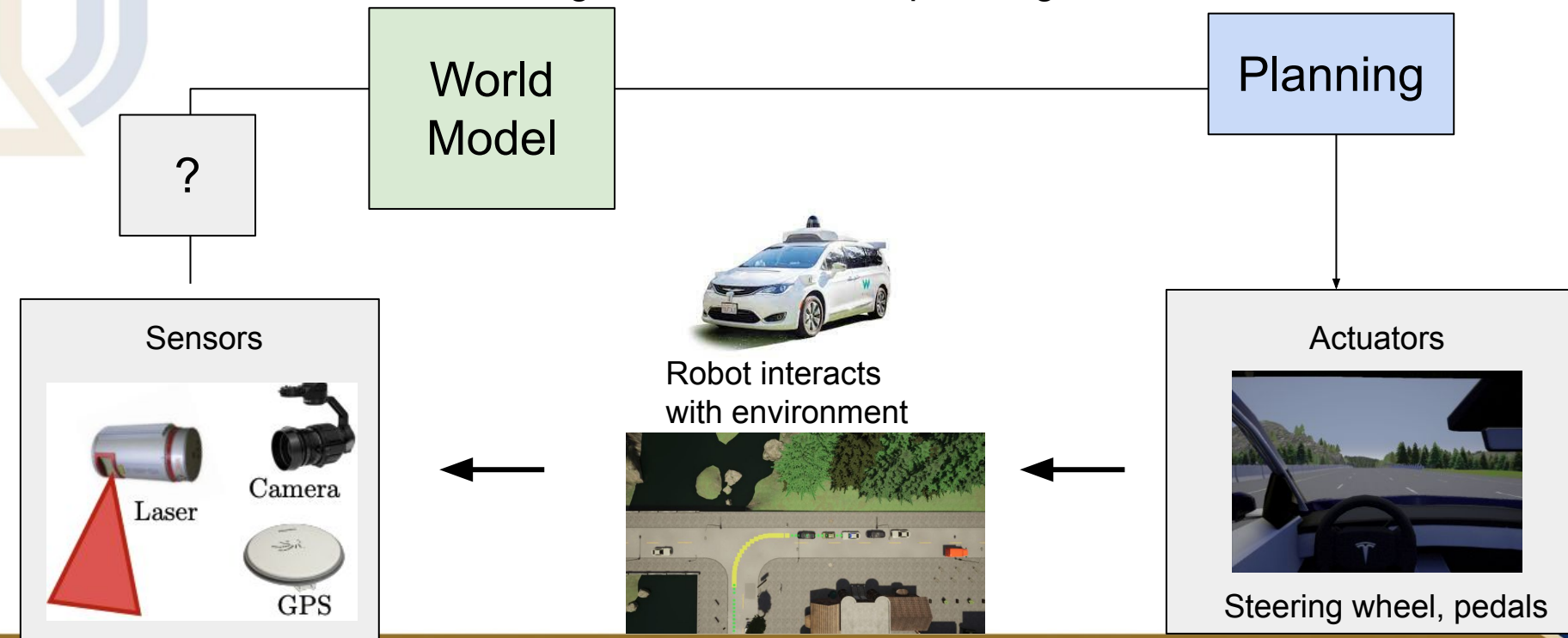
具身智能范式

- 有了世界知识，就可以做规划了！



The Embodied AI Paradigm

- Given the world knowledge, we can then do planning!



具身智能范式

- 什么是规划？

- 规划是一个优化问题：寻找最优动作序列，以最小化代价函数

- 在自动驾驶中，即最小化机器人到目标位置的距离

- 同时满足约束（不碰撞、遵守交通规则）

- 分为全局规划（全局路线）和局部规划（是否变道？转弯？）

The Embodied AI Paradigm

- What is planning?

- **Planning is an optimization problem in which we find the optimal sequence of actions towards minimizing a cost function**

- In the self-driving case, minimizing the robot's distance to the goal location

- While satisfying constraints (no crashing, follow traffic rules)

- There are global planning (global route) and local planning (change lane? Turn?)

具身智能范式

- 有了规划，控制呢？

规划给出要去哪里。现在需要控制机器人去执行

规划

控制

执行器



方向盘、踏板

11



机器人与环境交互



世界模型？

传感器



The Embodied AI Paradigm

- Given the plan, what about the control?

The plan gives where to go. Now we need to control the robot to do so

World Model

?

Planning

Control

Actuators



Steering wheel, pedals

11



Robot interacts with environment



Sensors



具身智能范式

- 什么是规划与控制？

- 分为全局规划（全局路线）和局部规划（是否变道？转弯？）
- 控制器通过向执行器发出具体指令来执行规划

■ 例如，将轨迹/运动规划（汽车应走哪条路、多快）转换为方向盘和踏板指令

The Embodied AI Paradigm

- What is planning and **control**?

- There are global (global route) and local planning (change lane? Turn?)
- **A controller will execute the plan by giving specific commands to the actuators**
 - For example, convert the trajectory/motion plan (which way the car should go and how fast) into the **steering** and **pedal** command

具身智能范式

- 关键问题

- Q1: 假设我们已知世界的一切。应向执行器发送什么指令?

- 已解决。如果了解世界, 我们就知道如何控制汽车

- Q2: 如何根据传感器更新世界知识?

The Embodied AI Paradigm

- Key questions

- Q1: Assume we know everything about the world. What commands should we send to the actuator?

- Done. We know how to control the car if we know about the world

- **Q2: How do we update the world knowledge given the sensors?**

具身智能范式

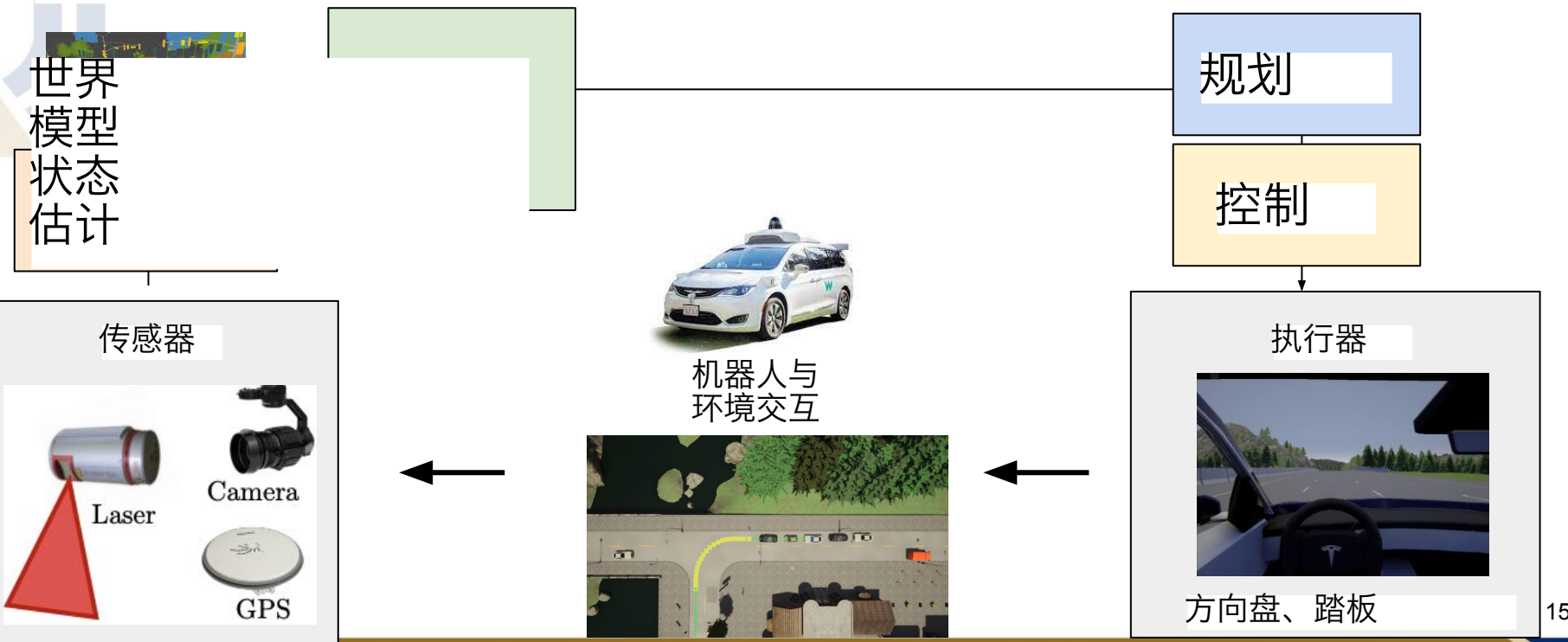
- 状态估计：根据传感器更新世界知识
 - 给定传感器数据，感知模块
 - 提取必要信息（关于周围环境）
 - 将机器人在世界中定位

The Embodied AI Paradigm

- **State Estimation** - Update the world knowledge given the sensors
 - Given the sensor data, the perception module
 - extracts necessary information (about the surroundings)
 - localizes the robot in the world

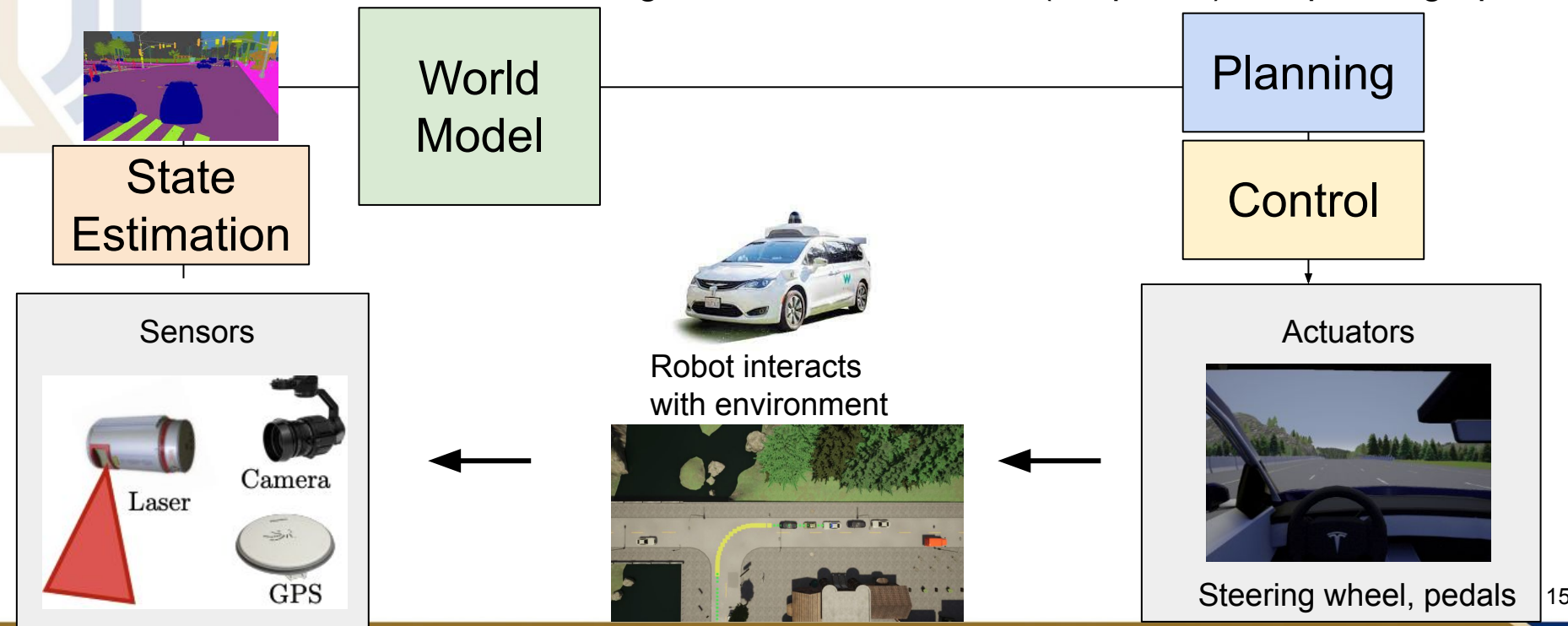
具身智能范式

- 如何打造自动驾驶汽车？——这是（简化后的）完整框图



The Embodied AI Paradigm

- How do we make a self-driving vehicle? - Here is the (simplified) completed graph



CMU 参加 2007 年 DARPA 自动驾驶挑战赛的冠军方案

- 来看一个成功的自动驾驶系统：来自 CMU 研究者



CMU's winning entry to DARPA Self-Driving challenge 2007

- Let's look at a successful self-driving system from researchers at CMU





Tartan Racing: A Multi-Modal Approach to the DARPA Urban Challenge

April 13, 2007

Chris Urmson, Joshua Anhalt, Drew Bagnell, Christopher Baker, Robert Bittner, John Dolan, Dave Duggins, Dave Ferguson, Tugrul Galatali, Chris Geyer, Michele Gittleman, Sam Harbaugh, Martial Hebert, Tom Howard, Alonzo Kelly, David Kohanbash, Maxim Likhachev, Nick Miller, Kevin Peterson, Raj Rajkumar, Paul Rybski, Bryan Salesky, Sebastian Scherer, Young Woo-Seo, Reid Simmons, Sanjiv Singh, Jarrod Snider, Anthony Stentz, William “Red” Whittaker, and Jason Ziglar

Carnegie Mellon University

Hong Bae, Bakhtiar Litkouhi, Jim Nickolaou, Varsha Sadekar, and Shuqing Zeng

General Motors

Joshua Struble and Michael Taylor

Caterpillar

Michael Darms

Continental AG

*最详尽记录的自动驾驶系统之一



Tartan Racing: A Multi-Modal Approach to the DARPA Urban Challenge

April 13, 2007

Chris Urmson, Joshua Anhalt, Drew Bagnell, Christopher Baker, Robert Bittner, John Dolan, Dave Duggins, Dave Ferguson, Tugrul Galatali, Chris Geyer, Michele Gittleman, Sam Harbaugh, Martial Hebert, Tom Howard, Alonzo Kelly, David Kohanbash, Maxim Likhachev, Nick Miller, Kevin Peterson, Raj Rajkumar, Paul Rybski, Bryan Salesky, Sebastian Scherer, Young Woo-Seo, Reid Simmons, Sanjiv Singh, Jarrod Snider, Anthony Stentz, William “Red” Whittaker, and Jason Ziglar

Carnegie Mellon University

Hong Bae, Bakhtiar Litkouhi, Jim Nickolaou, Varsha Sadekar, and Shuqing Zeng

General Motors

Joshua Struble and Michael Taylor

Caterpillar

Michael Darms

Continental AG

*One of the most well-documented autonomous driving system

CMU 参加 2007 年 DARPA 自动驾驶挑战赛的冠军方案

- 注意车顶传感器的数量
- 以及车辆建图与定位的 3D 可视化



18

CMU's winning entry to DARPA Self-Driving challenge 2007

- See the number of sensors on top of the vehicle
- And the 3D visualization of mapping and localization of the car



18

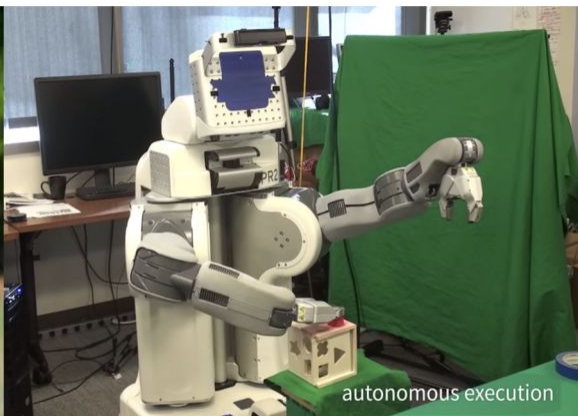
具身智能范式

- 除自动驾驶汽车外，该范式也可推广到其他机器人系统

○ 无人机、机械臂、腿式机器人



[Sa et al. IROS 2014]



[Levine et al. JMLR 2016]



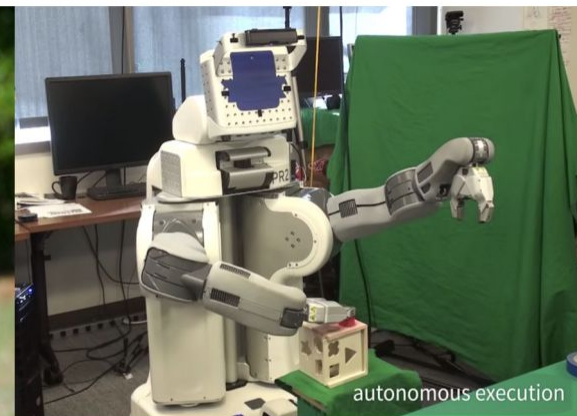
[Bohg et al. ICRA 2018]

The Embodied AI Paradigm

- Besides self-driving vehicle, we can also generalize this paradigm to other robot systems
 - Drones, robotic arms, legged machines



[Sa et al. IROS 2014]



[Levine et al. JMLR 2016]

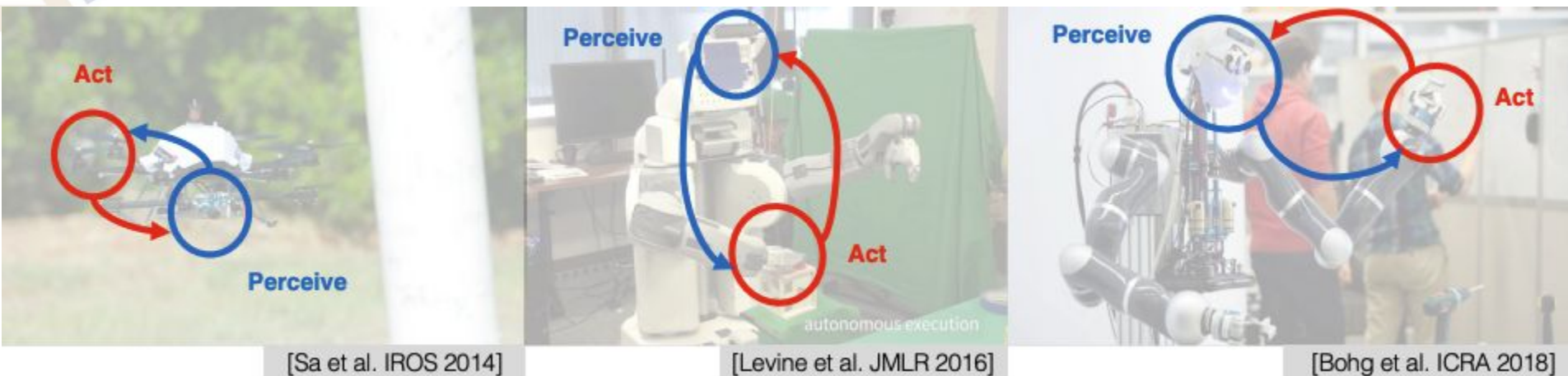


[Bohg et al. ICRA 2018]

具身智能范式

- 除自动驾驶汽车外，该范式也可推广到其他机器人系统

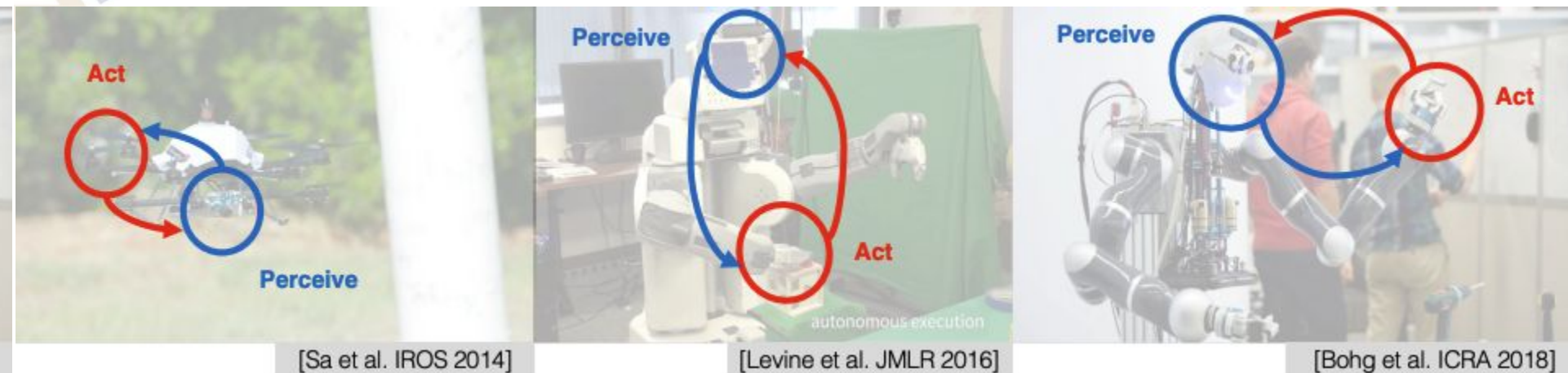
○ 关键是感知—动作闭环



感知 Perceive: 传感器类似: 相机、LiDAR、惯性测量单元 (IMU)
行动 Act: 执行器: 关节、机械臂、夹爪、电机

The Embodied AI Paradigm

- Besides self-driving vehicle, we can also generalize this paradigm to other robot systems
 - The key is the perception-action loop

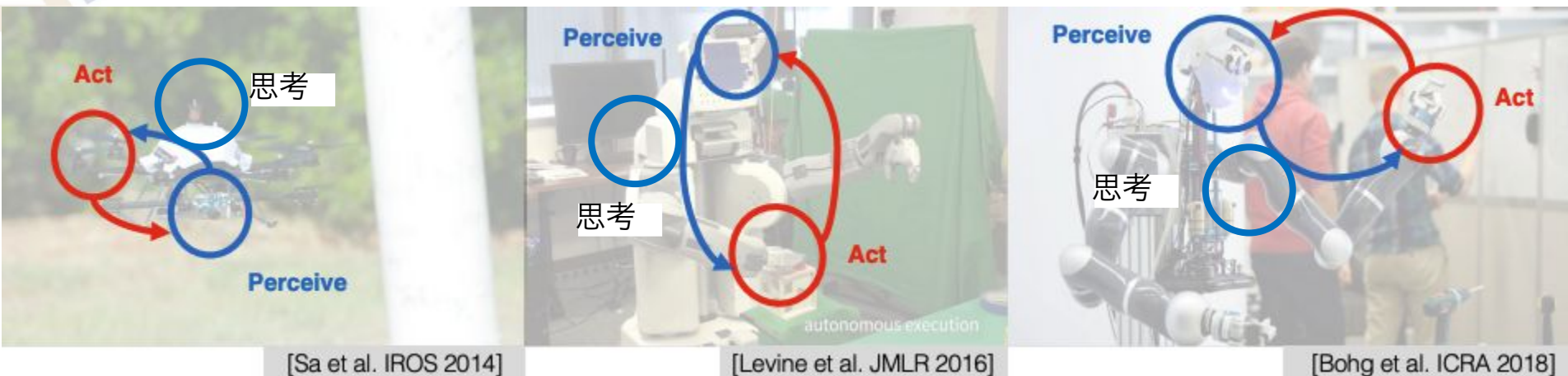


Perceive - Sensors are similar: Cameras, LIDARs, Inertial Measurement Units (IMUs)
Act - Actuators: Joints, arms, claws, motors

具身智能范式

- 除自动驾驶汽车外，该范式也可推广到其他机器人系统

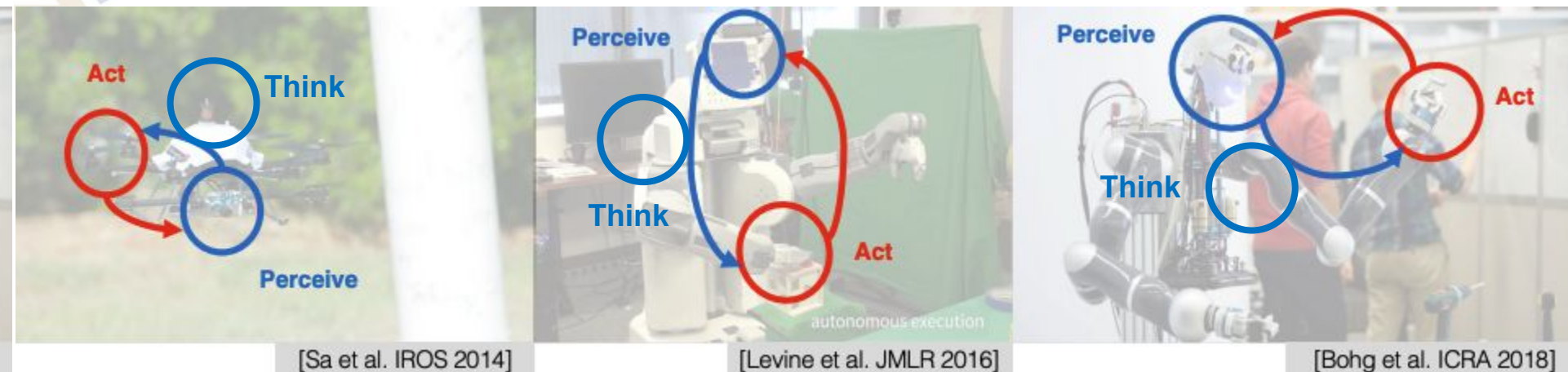
○ 加入机器学习以闭合感知—动作环（该过程称为“思考 Think”）



感知 Perceive: 传感器类似: 相机、LiDAR、惯性测量单元 (IMU)
行动 Act: 执行器: 关节、机械臂、夹爪、电机

The Embodied AI Paradigm

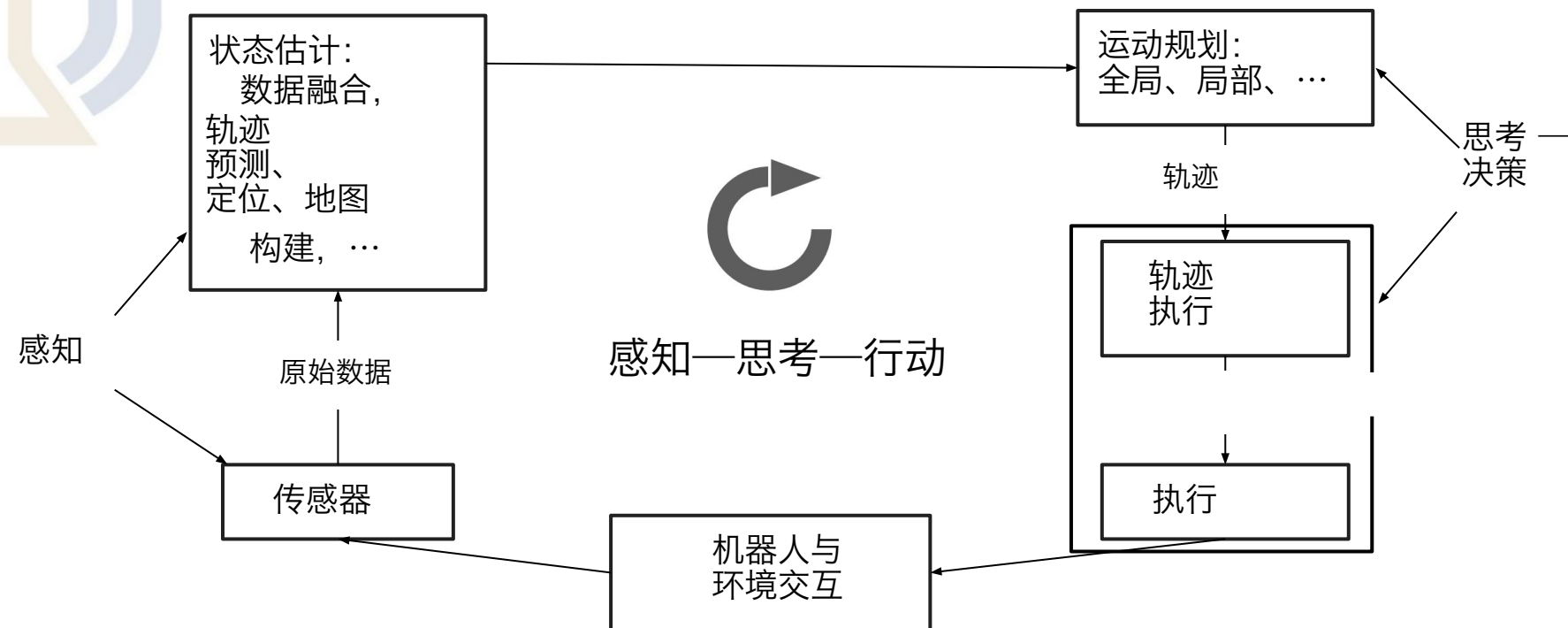
- Besides self-driving vehicle, we can also generalize this paradigm to other robot systems
 - Adding **machine learning** to close the perception-action loop (we call the process “Think”)



Perceive - Sensors are similar: Cameras, LIDARs, Inertial Measurement Units (IMUs)
Act - Actuators: Joints, arms, claws, motors

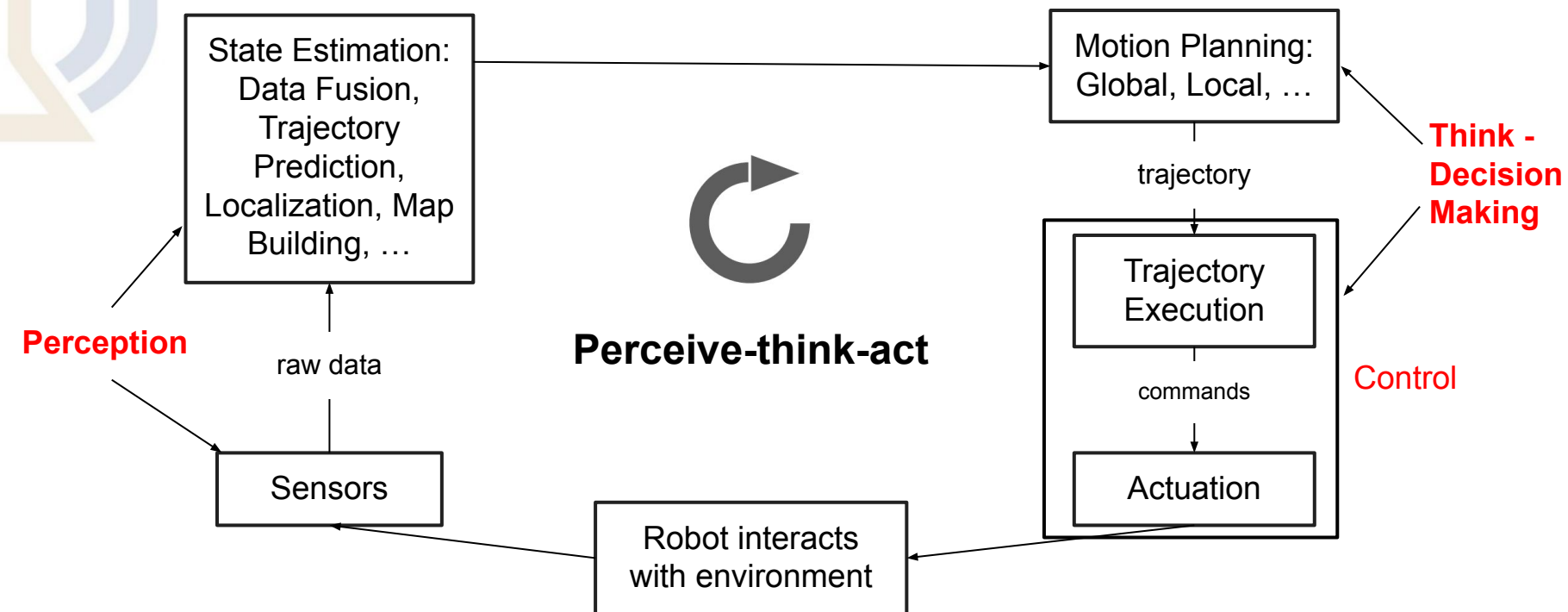
具身智能范式

- 更一般的范式——感知—思考—行动循环



The Embodied AI Paradigm

- A more generalized paradigm - The Perceive-think-act circle



前述自动驾驶系统中的感知—思考—行动循环

- 如何打造自动驾驶汽车？

世界模型
状态估计

传感器



机器人与环境交互



规划

控制

行动

执行器



方向盘、踏板

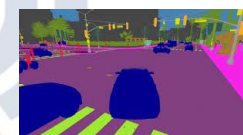
23

The Perceive-Think-Act Cycle in previous self-driving system

Perceive

do we make a self-driving vehicle?

Think



State Estimation

World Model

Planning

Control

Act

Actuators



Sensors



Robot interacts with environment



Steering wheel, pedals

23

目录

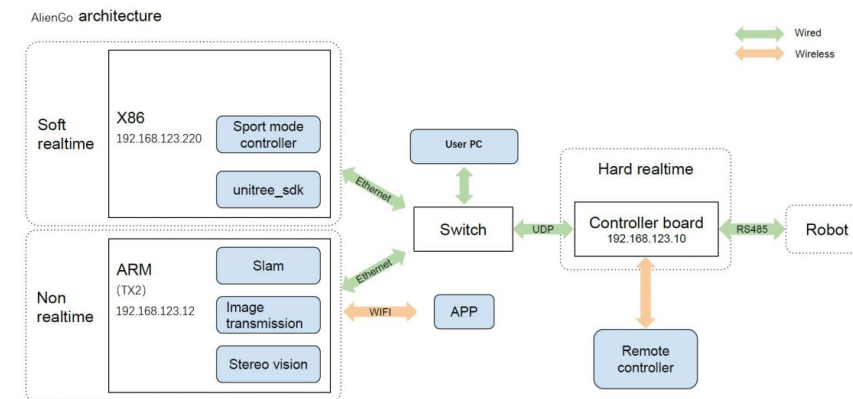
- 具身智能范式
- 具身系统设计
- 传感器概览
- 机器人视觉 vs. 计算机视觉

Agenda

- The Embodied AI paradigm
- **Embodied System Design**
- Sensors Overview
- Robot Vision vs. Computer Vision

具身系统设计

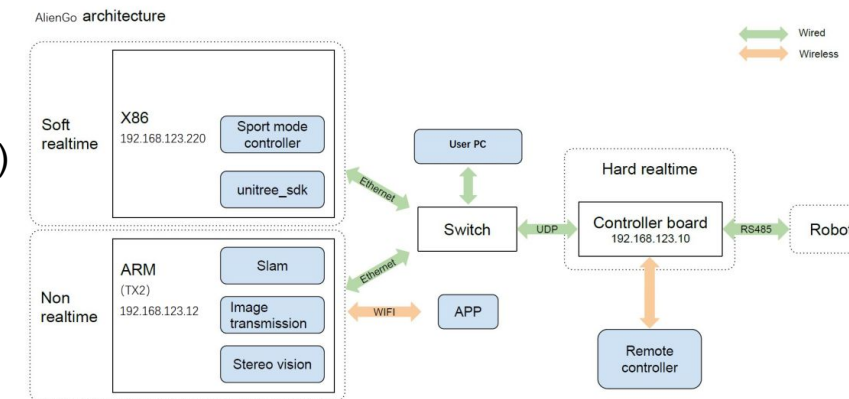
- 硬件
 - 传感器
 - 执行器
- 软件
 - 底层：控制单元驱动
 - 中间件：机器人操作系统（ROS）
- 基于数据分发服务（DDS）
- 模块间通信
- 数据记录
- 可视化
- 感知模型
- 运动规划模型
- 原则
 - 遵循感知—思考—行动范式
 - 模块化、鲁棒、数据记录与回放



宇树（Unitree）
四足机器人的架构

Embodied System Design

- The hardware
 - Sensors
 - Actuators
- The software
 - Lowlevel: Control Unit Drivers
 - Middleware: Robot Operating System (ROS)
 - Based on Data Distribution Service (DDS)
 - Communication between modules
 - Data logging
 - Visualization
 - Perception models
 - Motion planning models
- Principles
 - Follow the Perceive-think-act paradigm
 - Modular, robust, data logging and playback

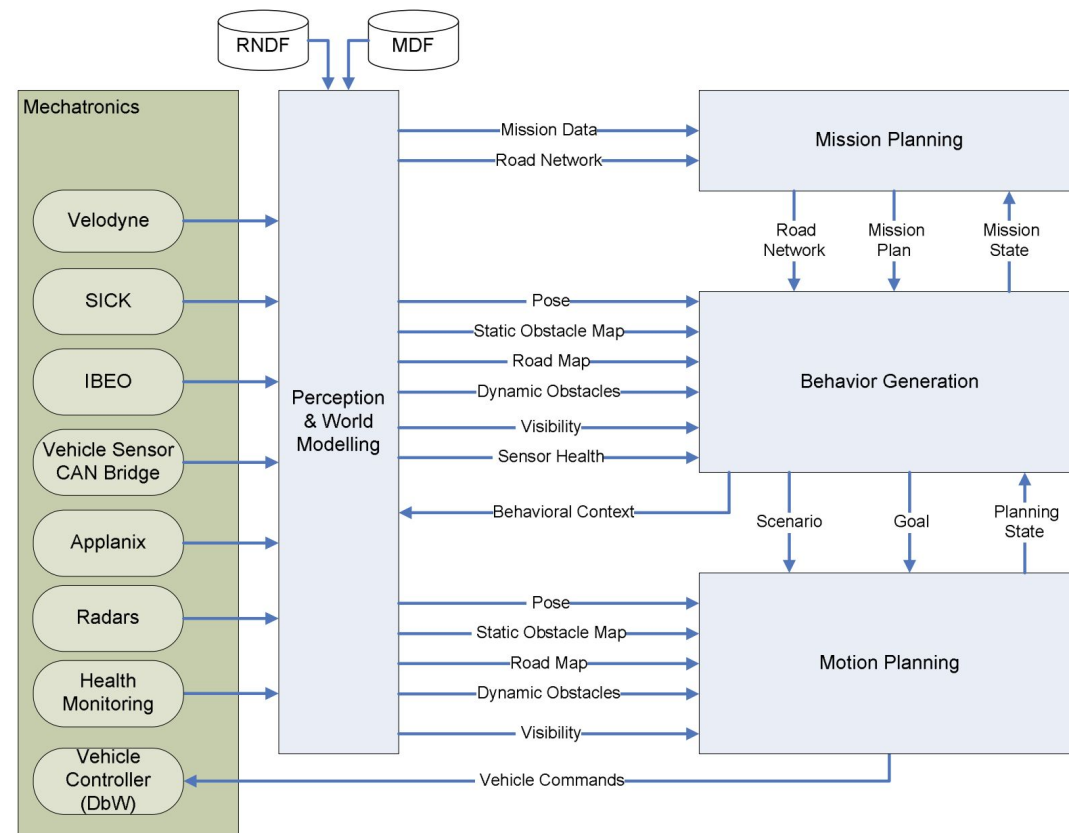


The architecture of Unitree's
four legged-robot

具身系统设计

• 示例: CMU 的自动驾驶系统

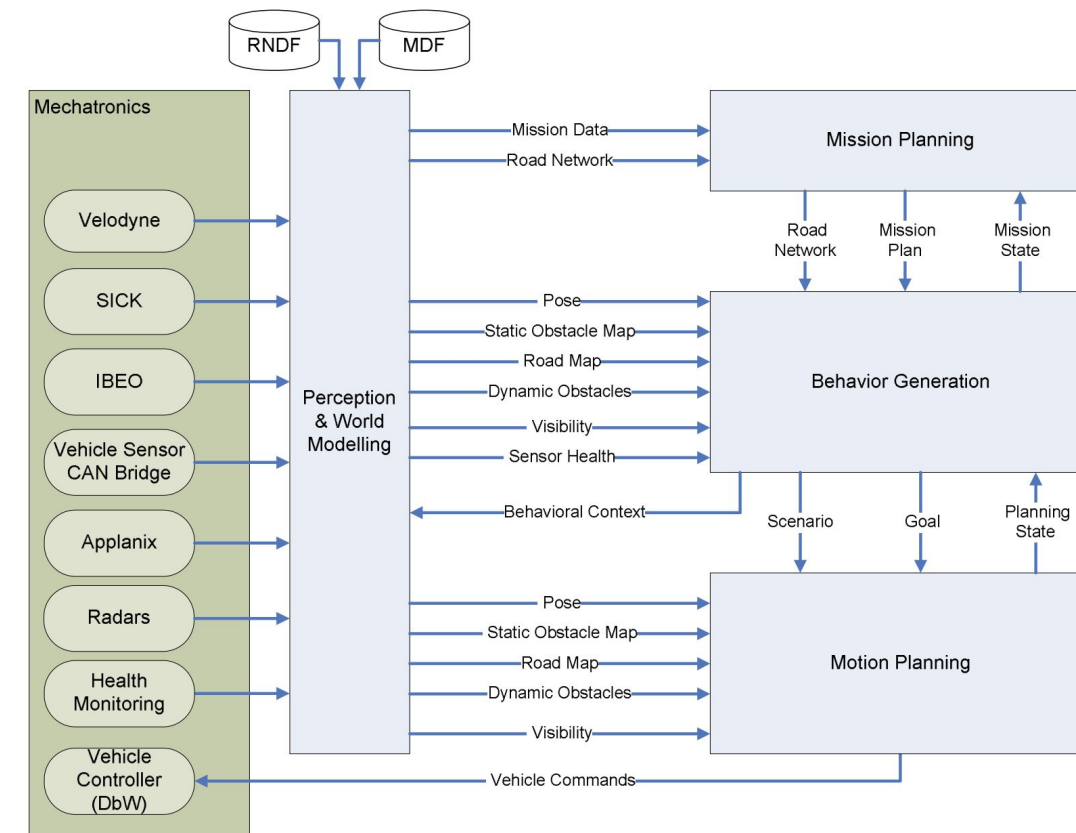
- 感知
- 规划
- 机电一体化
(控制器与传感器)



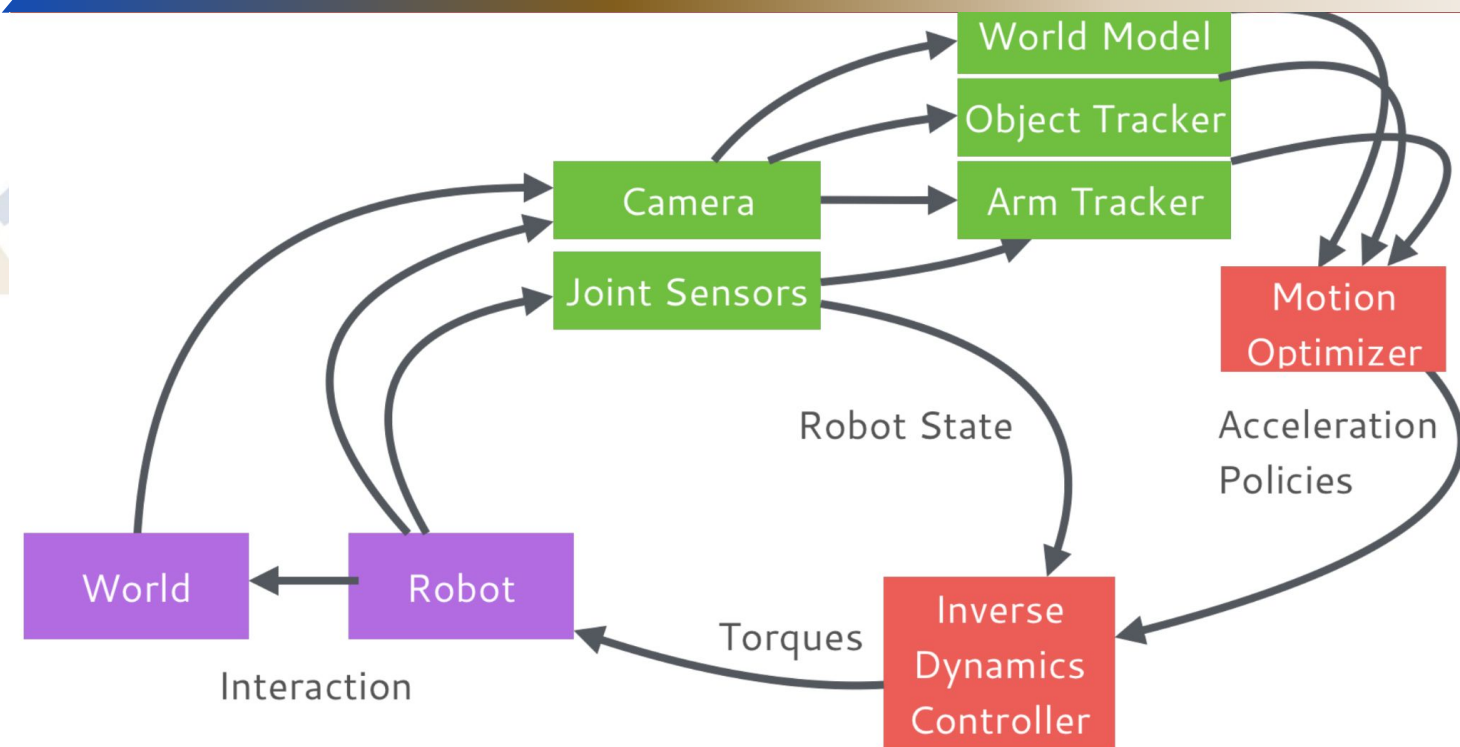
Embodied System Design

• Example: CMU's Self-driving system

- Perception
- Planning
- Mechatronics
(Controller & sensors)

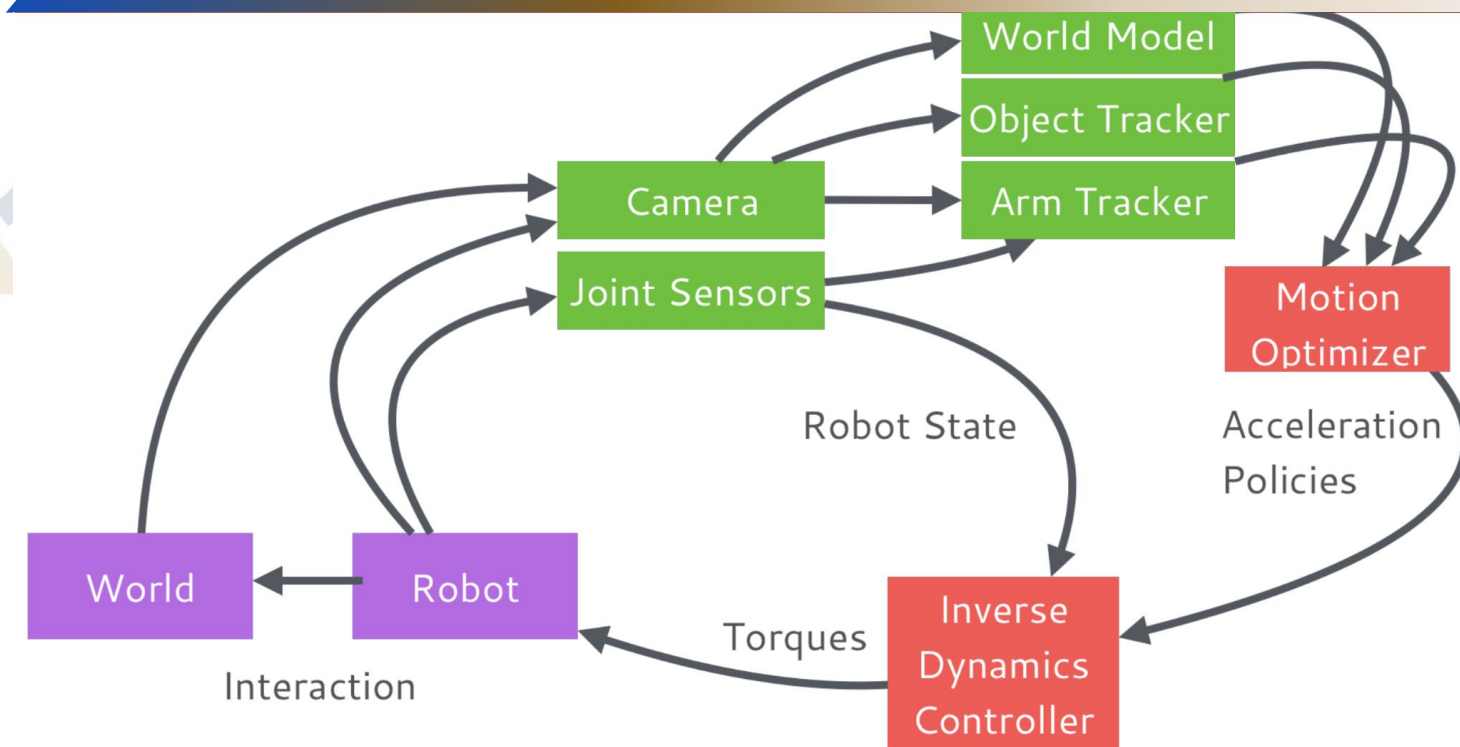


具身系统设计



• 示例: Kappler et. al. Real-Time Perception Meets Reactive Motion Generation. ICRA 2018

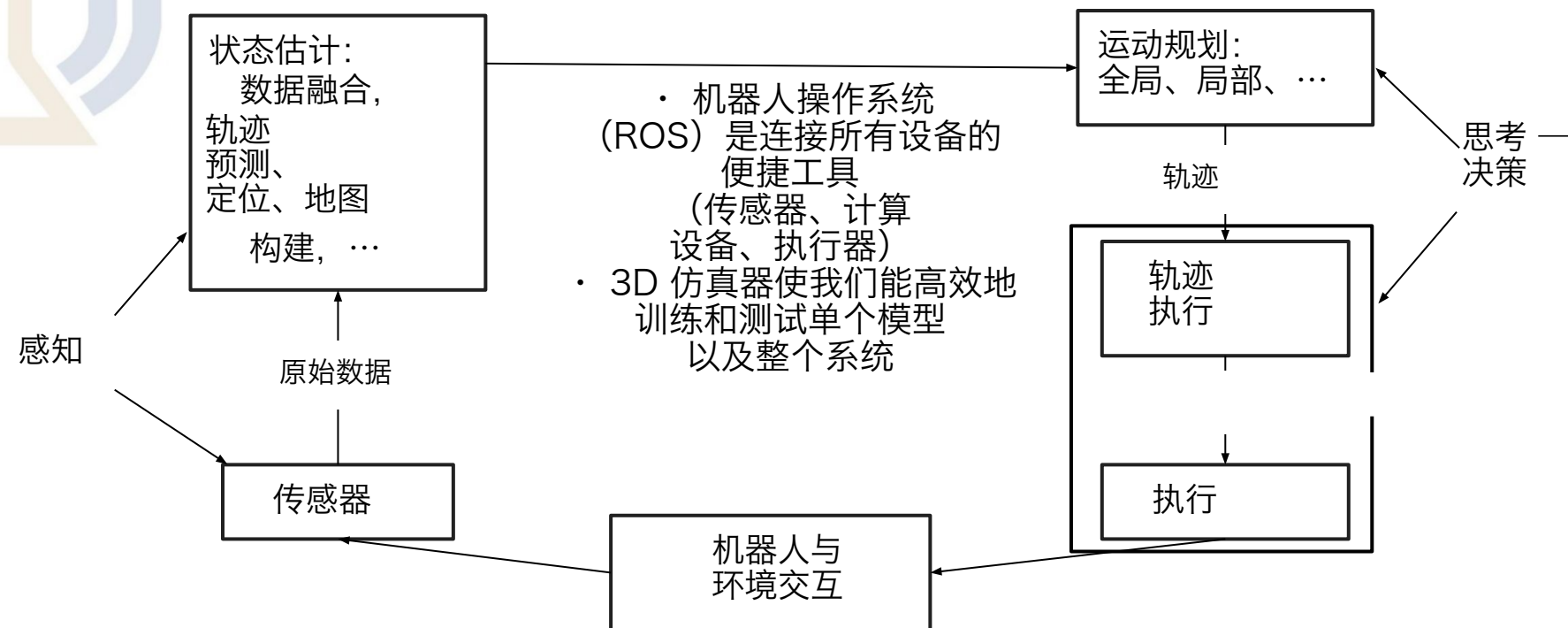
Embodied System Design



• Example: Kappler et. al. Real-Time Perception Meets Reactive Motion Generation. ICRA 2018

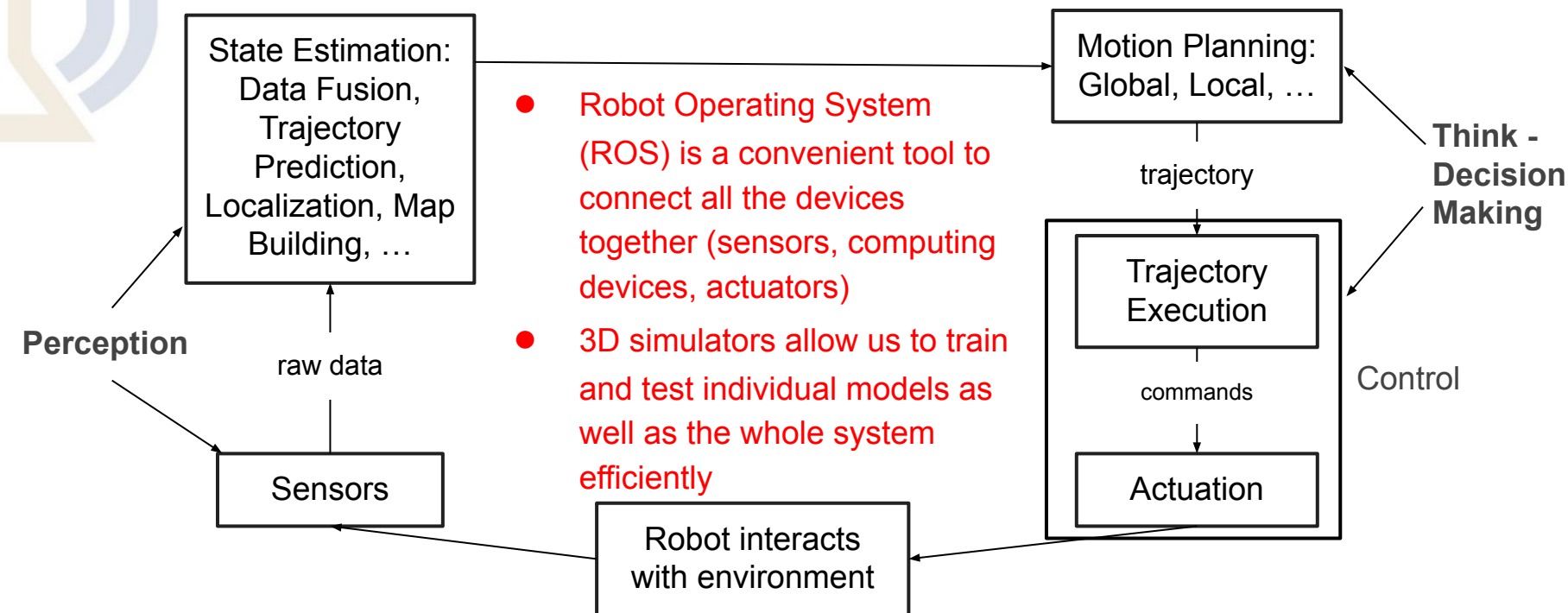
具身系统与具身智能范式

- ROS 与 3D 仿真器是具身智能的关键组件



Embodied System and The Embodied AI Paradigm

- ROS and 3D Simulator are key component for Embodied AI



目录

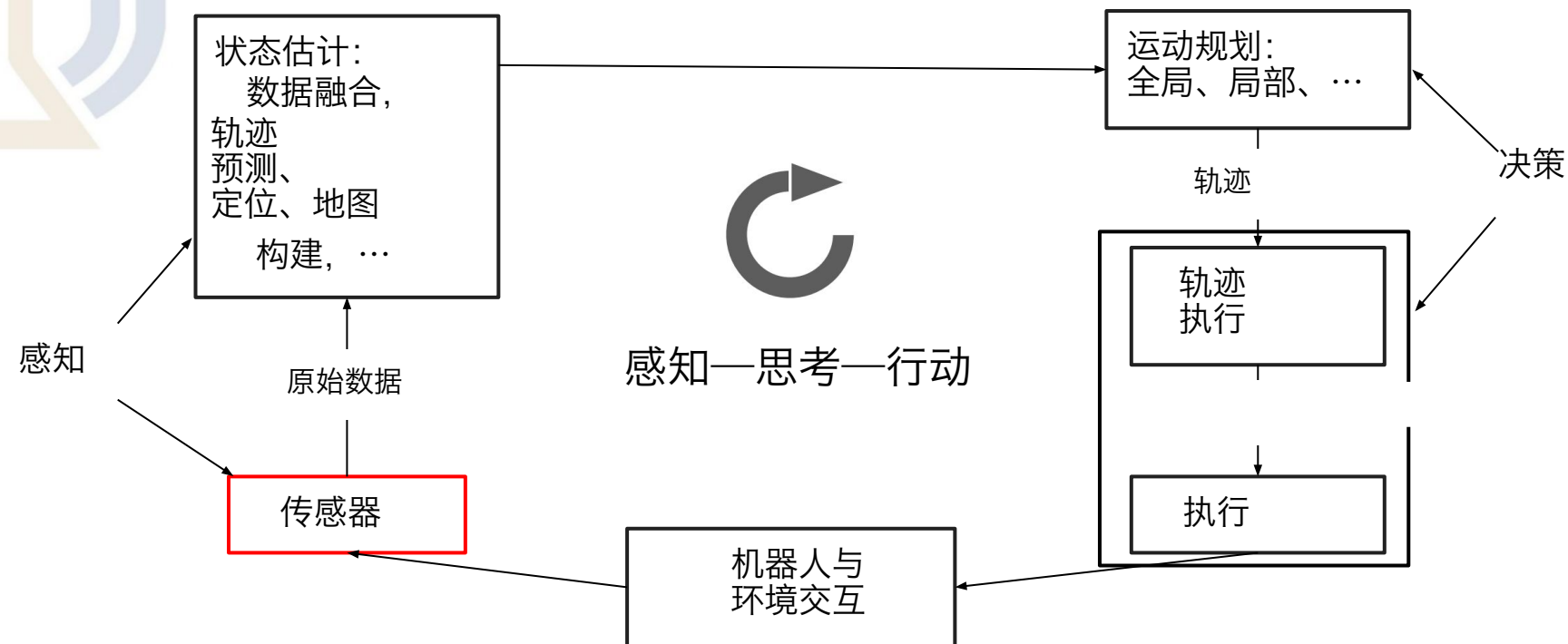
- 具身智能范式
- 具身系统设计
- 传感器概览
- 机器人视觉 vs. 计算机视觉

Agenda

- The Embodied AI paradigm
- Embodied System Design
- **Sensors Overview**
- Robot Vision vs. Computer Vision

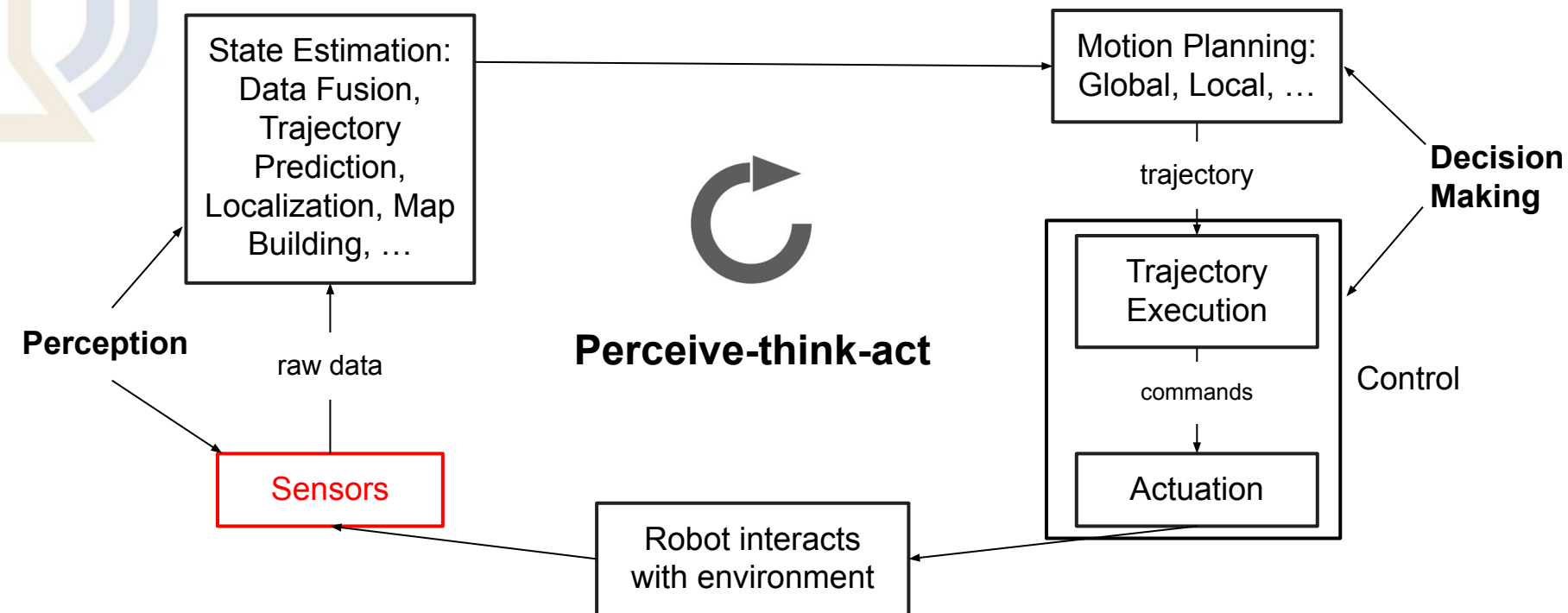
具身智能范式

- 感知—思考—行动循环



The Embodied AI Paradigm

- The Perceive-think-act circle

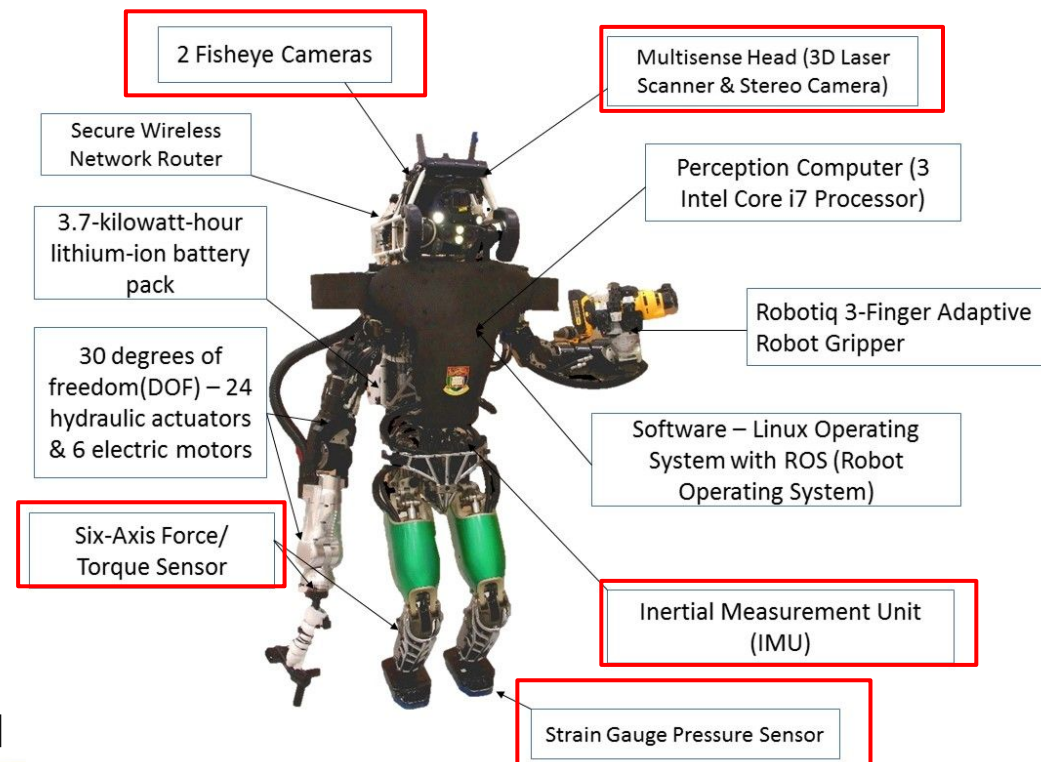


机器人传感器

- 多模态传感器是机器人感知的关键

○ 示例：2015 年 DARPA 机器人挑战赛

- 鱼眼相机
- LiDAR
- 惯性测量单元 (IMU)
- 力矩传感器
- 压力传感器



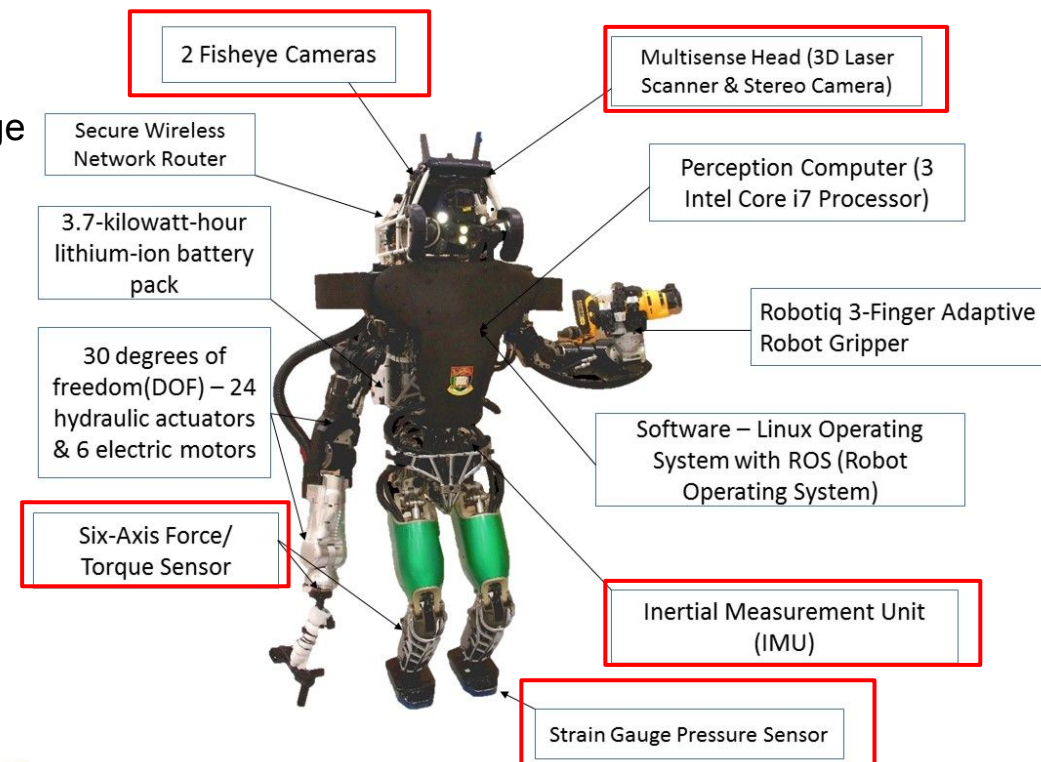
[来源：HKU Advanced Robotics Laboratory]

Robotic Sensors

- Multimodal sensors are key to robot perception

○ Example: DARPA robotic challenge 2015

- Fisheye cameras
- LiDAR
- Inertial Measurement Unit (IMU)
- Torque Sensor
- Pressure Sensor



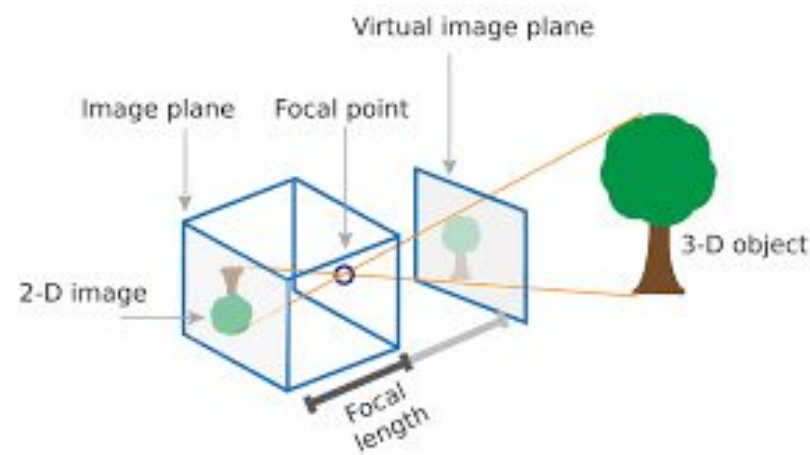
[Source: HKU Advanced Robotics Laboratory]

机器人传感器

- 多模态传感器是机器人感知的关键

- 相机、立体相机

- 三维世界如何投影到二维图像?
- 我们将在第4讲学习!

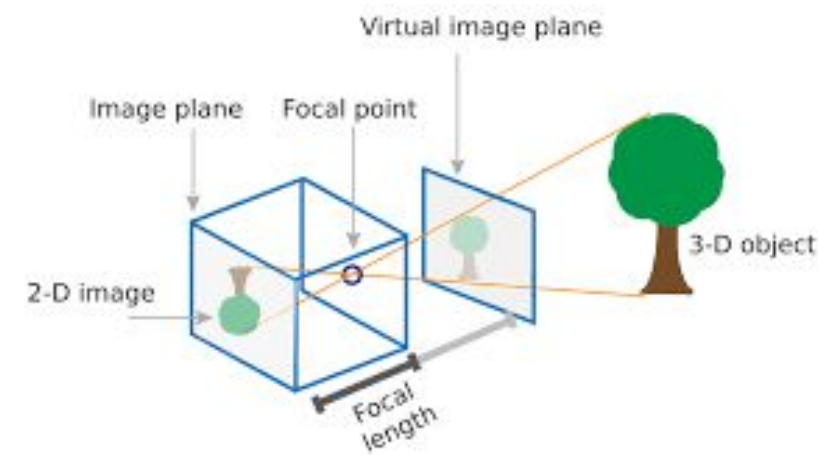


Robotic Sensors

- Multimodal sensors are key to robot perception

- Cameras, Stereo Cameras

- How does the 3D world project to a 2D image?
- We will learn about it in lecture 4!

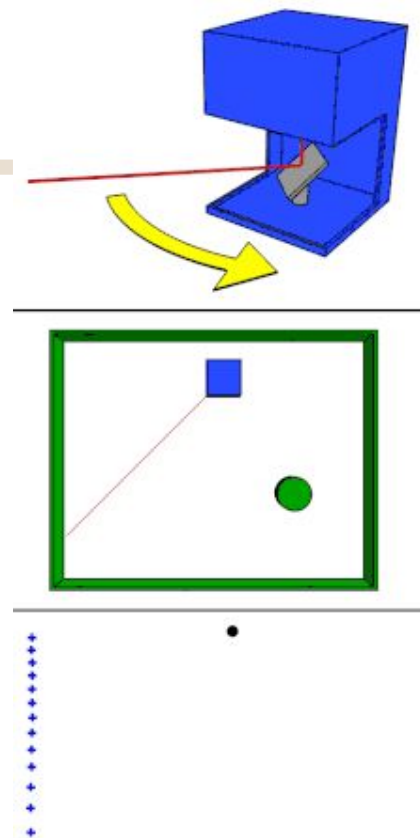


机器人传感器

- 多模态传感器是机器人感知的关键

○ LiDAR

- light detection and ranging (光探测与测距) 的缩写
- 发射激光并获取到障碍物的距离
- 为什么会有不连续?
 - 只能对有限角度进行采样



Robotic Sensors

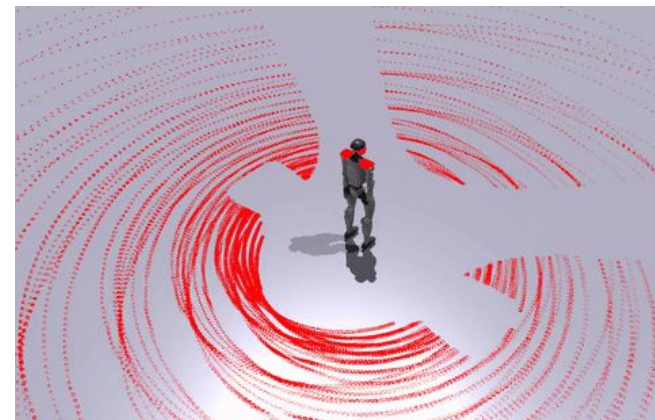
- Multimodal sensors are key to robot perception

○ Lidar

- Short for light detection and ranging
- Send laser and get distance to obstacles
- Why the discontinuance?
 - You can only sample finite angles



LiDAR Sim from CARLA



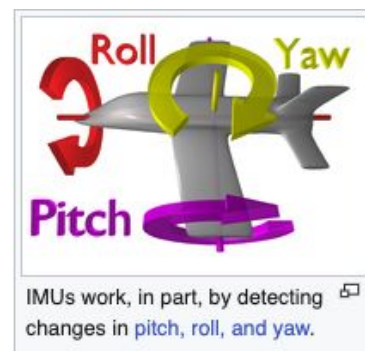
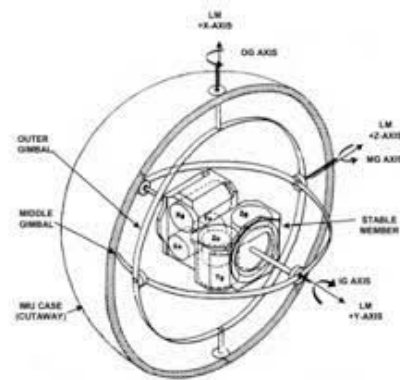
LiDAR Sim on robot from our work: OmniPerception-CoRL'25

机器人传感器

- 多模态传感器是机器人感知的关键

- 惯性测量单元 (IMU)

- 测量角速度、姿态、加速度等
- 用于无人机和智能手机!

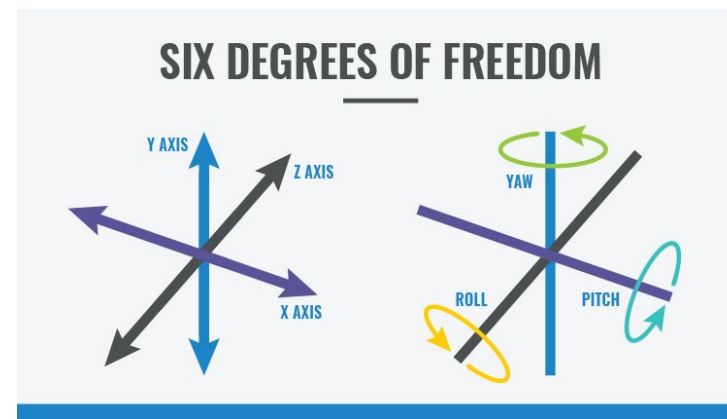
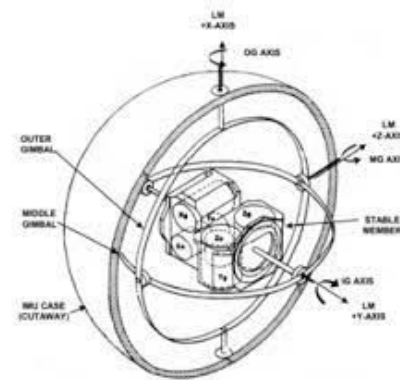


Robotic Sensors

- Multimodal sensors are key to robot perception

- Inertial Measurement Unit (IMU)

- Measure angular rate, orientation, acceleration, etc.
- Used in drones, and smartphones!

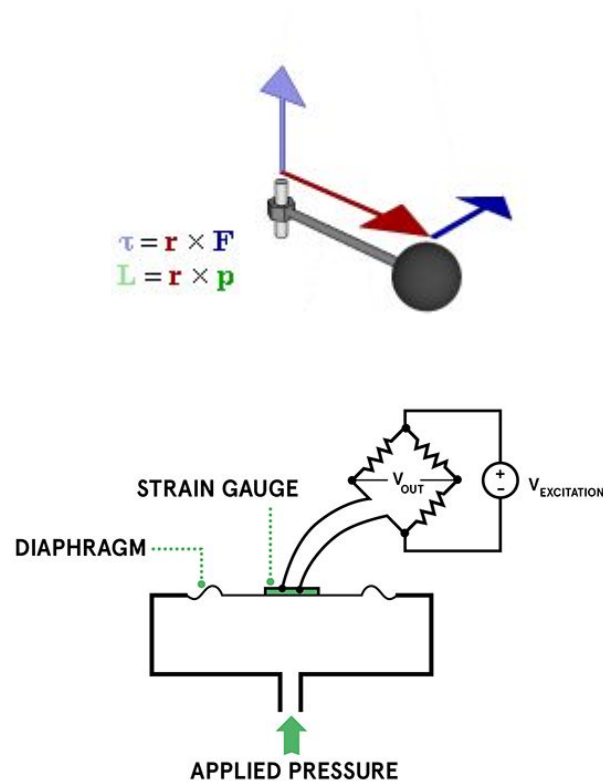


机器人传感器

• 多模态传感器是机器人感知的关键

- 力矩传感器
- 测量旋转力
- 应变片压力传感器

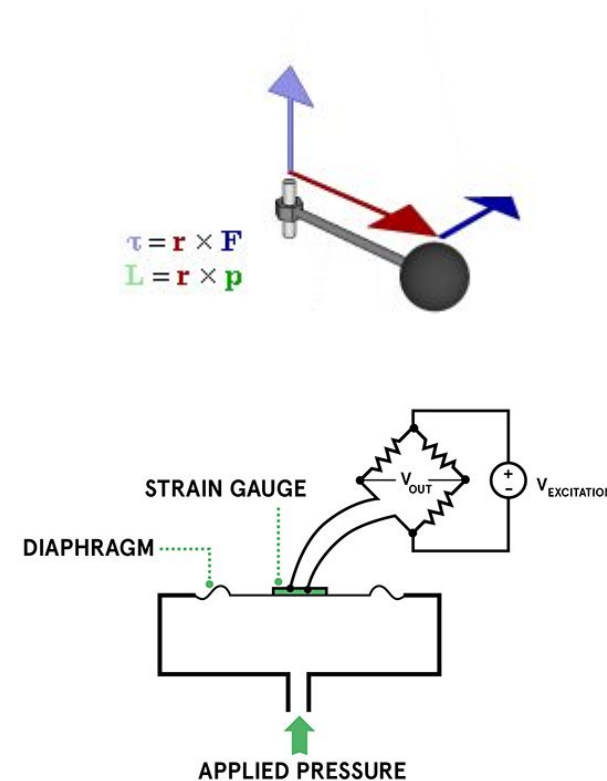
■ 施加压力，电信号随之变化
■ 触觉传感器！



Robotic Sensors

• Multimodal sensors are key to robot perception

- Torque Sensor
 - Measure rotation force
- Strain Gauge Pressure Sensor
 - Apply pressure and electronic signal changes
 - Touch sensor!

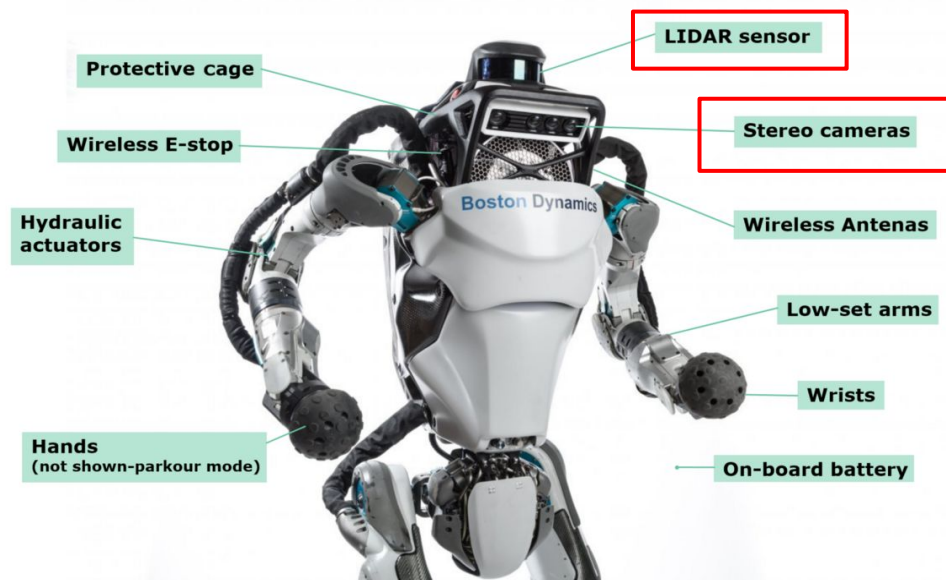


机器人传感器

- 多模态传感器是机器人感知的关键

○ Boston Dynamics Atlas

- 相机
- LiDAR
- 惯性测量单元 (IMU) ?

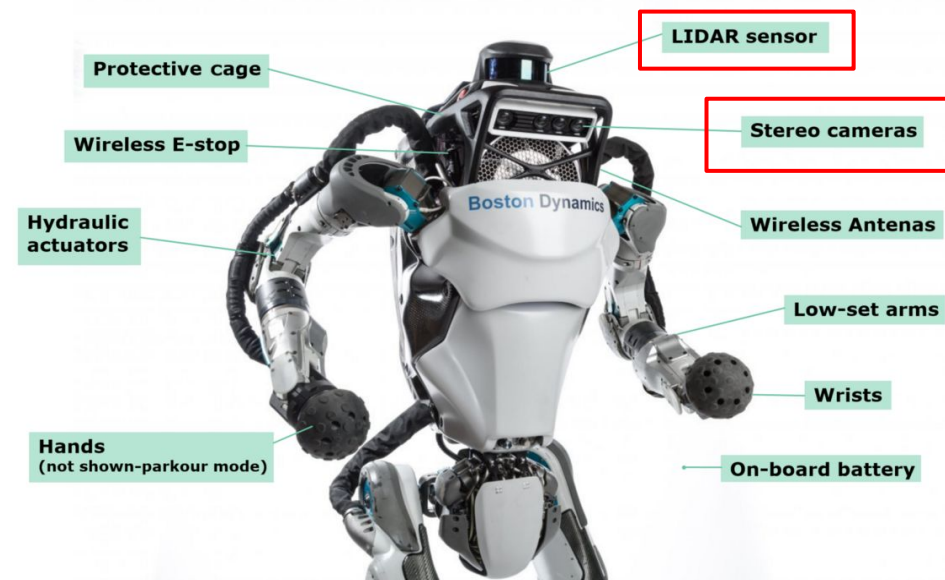


Robotic Sensors

- Multimodal sensors are key to robot perception

○ Boston Dynamics Atlas

- Cameras
- Lidar
- Inertial Measurement Unit (IMU)?



机器人传感器

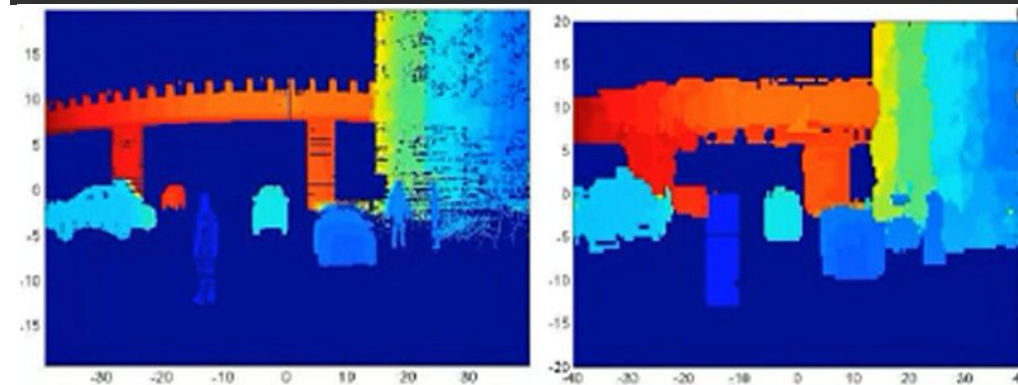
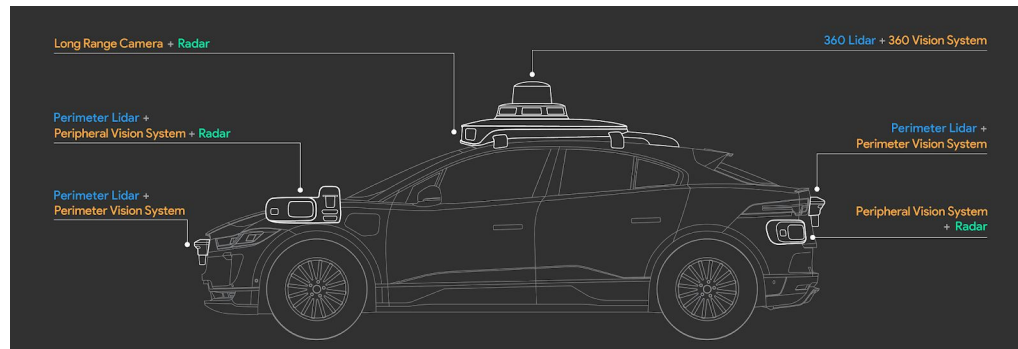
- 多模态传感器是机器人感知的关键

○ Waymo 自动驾驶汽车

- 相机
- LiDAR

- 使用近红外
- 更精确
- 雷达

- 使用微波
- 距离更远



Lidar

High Resolution Radar

Robotic Sensors

- Multimodal sensors are key to robot perception

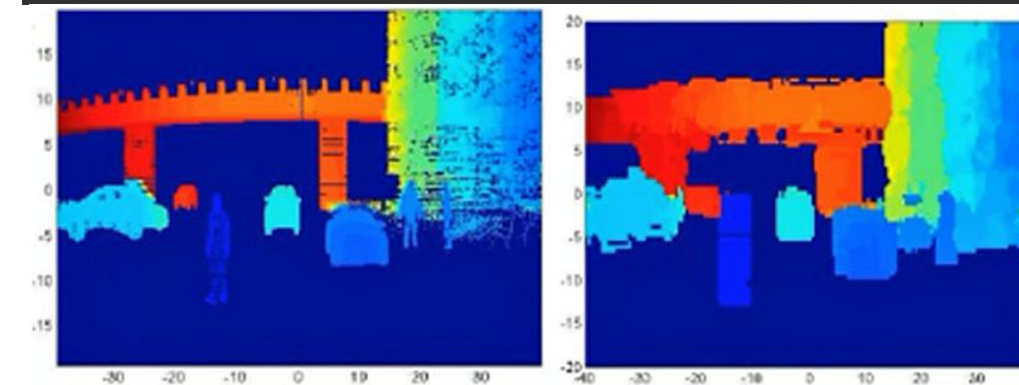
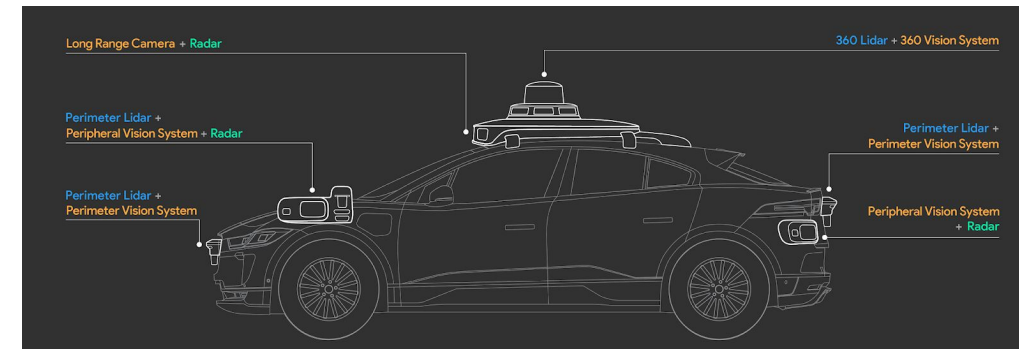
○ Waymo Self-driving Cars

- Cameras
- Lidar

- Uses near-infrared
- More precise

- Radar

- Uses microwave
- Longer range



Lidar

High Resolution Radar

机器人传感器

- 多模态传感器是机器人感知的关键
 - 在 3D 仿真器 CARLA 中
 - RGB 相机、IMU、深度相机、GPS、雷达、LiDAR（可设置 LiDAR 属性）、事件相机等
 - 事件相机：基本无限 FPS 的（低分辨率）相机



Robotic Sensors

- Multimodal sensors are key to robot perception
 - In 3D simulator CARLA
 - RGB Camera, IMU, Depth Camera, GPS, Radar, Lidar (you can set Lidar properties), Event Camera, etc.
 - Event camera: basically unlimited FPS (low-resolution) camera



目录

- 具身智能范式
- 具身系统设计
- 传感器概览
- 机器人视觉 vs. 计算机视觉

Agenda

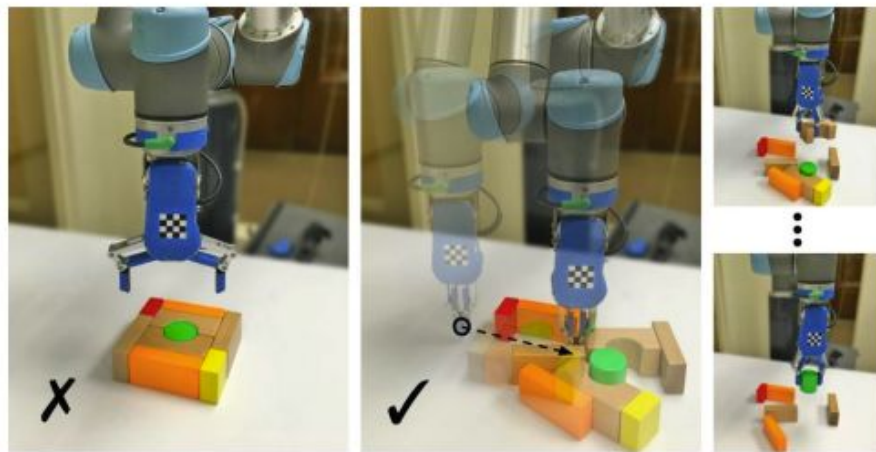
- The Embodied AI paradigm
- Embodied System Design
- Sensors Overview
- Robot Vision vs. Computer Vision

机器人视觉 vs. 计算机视觉

- 机器人视觉是具身的、主动的、情境化的



[Detectron - Facebook AI Research]



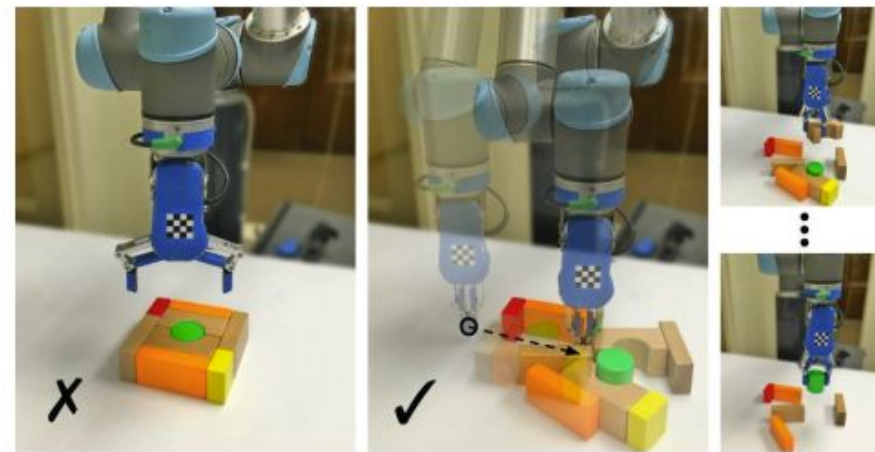
[Zeng et al., IROS 2018]

Robot Vision vs. Computer Vision

- Robot vision is embodied, active and environmentally situated



[Detectron - Facebook AI Research]



[Zeng et al., IROS 2018]

机器人视觉 vs. 计算机视觉

- 机器人视觉是具身的、主动的、情境化的
- 具身：机器人拥有物理身体，直接体验世界。它们的动作与环境共同构成动力学系统，并即时反馈到自身的感觉
- 主动：机器人是主动感知者。它们应知道感知什么、为何感知，并能选择感知内容，以及决定如何、何时、何地完成该感知
- 情境化：机器人处于世界之中。环境会影响整个系统

Robot Vision vs. Computer Vision

- Robot vision is **embodied, active** and **environmentally situated**
- **Embodied**: Robots have physical bodies and experience the world directly. Their actions are part of a dynamical system together with the environment and have **immediate feedback** on their own sensation
- **Active**: Robots are **active perceivers**. They should know what and why it wishes to sense, and they should be able to choose what to perceive, and determine how, when and where to achieve that perception
- **Situated**: Robots are situated in the world. The environment will affect the whole system

机器人视觉 vs. 计算机视觉

- 机器人视觉是具身的、主动的、情境化的



Robot Vision vs. Computer Vision

- Robot vision is **embodied, active** and **environmentally situated**
 - Like human ego-vision



目录

- 具身智能范式
- 具身系统设计
- 传感器概览
- 机器人视觉 vs. 计算机视觉
- 实验室近期机器人演示

Agenda

- The Embodied AI paradigm
- Embodied System Design
- Sensors Overview
- Robot Vision vs. Computer Vision
- Recent robot demos from our lab

具身智能实验室 / Precognition Lab Overview



人员与资助、项目与实验室

- PI: Junwei Liang, 助理教授, CMU 博士
- CoRL/NeurIPS Area Chair, CoRL GenPriors Best Paper
- 项目: NSFC、腾讯 Robotics X、美团机器人、阿里巴巴、华为等
- 总额 500 万+ 人民币
- 实验室: 200+ 平方米机器人实验空间
- 10 名博士生。即将毕业者拿到 300 - 900 万人民币 offer

硬件

- 共有 80 块 GPU
- A6000/Pro 6000 GPU 服务器
- L40/4090 工作站
- 机器人平台
 - 人形机器人 (n* Unitree G1 与 H2)
 - 四足机器人 (A2 与 Go2-W)

具身智能实验室 / Precognition Lab Overview

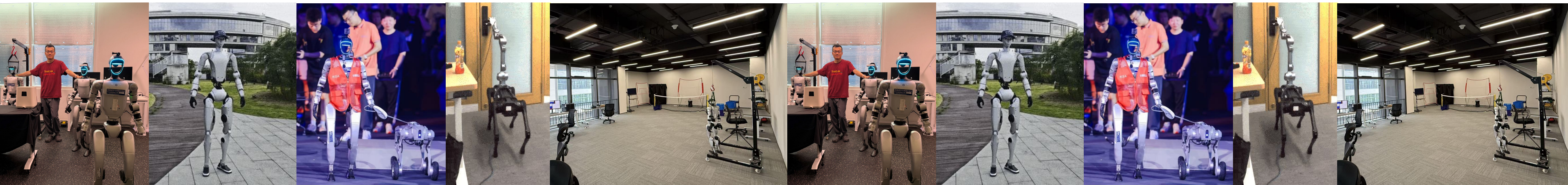


People & Funding Projects & Lab

- **PI: Junwei Liang, Asst. Prof., CMU PhD.**
 - CoRL/NeurIPS Area Chair, CoRL GenPriors Best Paper
- Projects: NSFC, Tencent Robotics X, Meituan Robotics, Alibaba, Huawei, etc.
 - **Totaling 5M+ RMB**
- Lab: **200+ Sq Meters lab** for robot experiments
- 10 PhD students. **Soon-graduates got 3-9M RMB offer**

Hardware

- We have a total of 80 GPUs
 - A6000/**Pro 6000** GPU servers
 - L40/4090 workstations
- **Robot Platforms**
 - Humanoid robots (n* Unitree G1 and H2)
 - Four-legged robots (A2 and Go2-W)



具身智能实验室 / Precognition Lab - Overview

- 目标：构建人类水平的 AI 系统，使其能够有效感知、推理并与现实世界交互，造福人类
 - 最终得到能帮助我们完成日常任务的通用家用机器人

空间智能 ☒ VLM

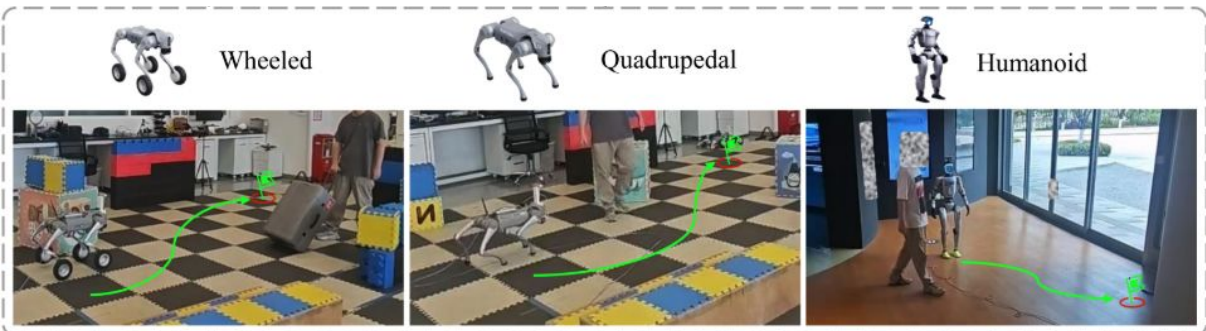
“行动”

移动能力：导航 + 运动控制

操作

“如何做事”

“如何到达某处”



实验室网站: <https://precognition.team/>

具身智能实验室 / Precognition Lab - Overview

- Goal: Build human-level AI systems that could effectively perceive, reason and interact with the real-world for the good of humans
 - So eventually we have a **general household robot** that can help us with everyday tasks

Spatial Intelligence → VLM

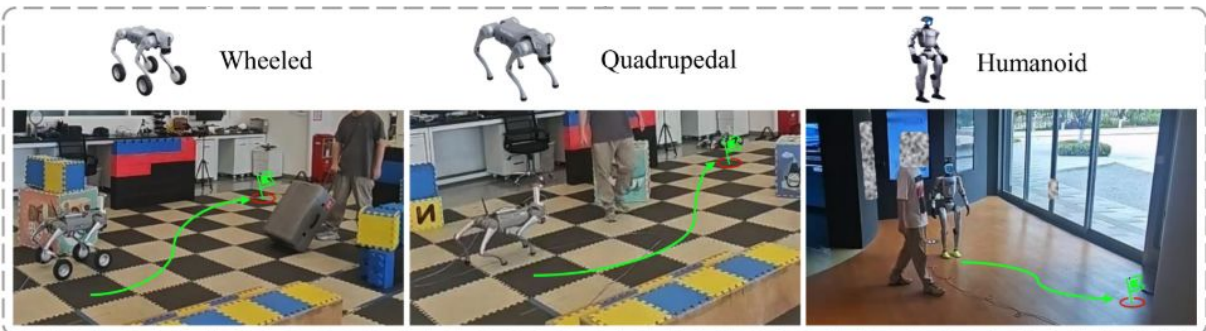
“Action”

Mobility - Navigation + Locomotion

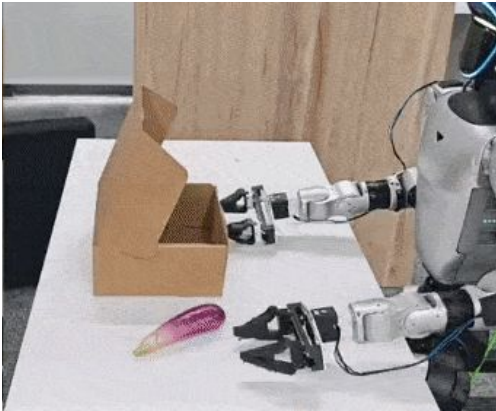
Manipulation

“How to do stuff”

“How to get somewhere”



Lab website: <https://precognition.team/>



Precognition Lab 研究亮点

空间智能 ☒ VLM

“行动”

移动：导航/运动控制

“在何处做事”

操作

“如何做事”

• 迈向通用服务具身智能

- 三维视觉定位 - SeeGround (CVPR’ 25, NeurIPS’ 25)
- 社交导航 - FLUX (*ICRA’ 25, IROS’ 26, RA-L’ 26)
- 基于 LiDAR 的 WBC 运动控制 (CoRL’ 25 Oral)
- 感知行为基础模型 (CoRL’ 26)
- 操作模仿学习 - GLOVER (CoRL’ 25)
- VLA 操作基准 - AGNOSTOS (NeurIPS’ 25)
 - 运动世界动作模型 (*’ 26)
- 端到端 SpeechVLA 人形智能体 (*’ 26 已部署于大学访客中心)

*今天展示这些演示

Research Highlights of Precognition Lab

Spatial Intelligence → VLM

“Action”

Mobility: Navigation/Locomotion

“Where to Do stuff”

Manipulation

“How to Do stuff”

Towards General Service Embodied AI

- 3D Visual Grounding - SeeGround (CVPR’25, NeurIPS’25)
- **Social Navigation - FLUX (*ICRA’25, IROS’26, RA-L’26)**
- LiDAR-based WBC Locomotion (CoRL’25 Oral)
- **Perceptive Behavior Foundation Model (CoRL’26)**
- Imitation learning for Manipulation - GLOVER (CoRL’25)
- VLA Manipulation Benchmark - AGNOSTOS (NeurIPS’25)
- **Motion World Action Model (*’26)**
- **End-to-end SpeechVLA Humanoid Agent (*’26 Deployed at Uni. Visitor Center)**

[*We show these demo today](#)

From Cognition to Precognition: A Future-Aware Framework for Social Navigation 基于未来预测的社交导航系统

1 AI Thrust, 香港科技大学 (广州)
2 计算机科学与工程系, 香港

科技大学
*通讯作者

所有作者都是我的学生,
我是唯一导师。
工作耗时约 8 个月。

<https://zeying-gong.github.io/projects/falcon/>

From Cognition to Precognition: A Future-Aware Framework for Social Navigation 基于未来预测的社交导航系统

Zeying Gong¹, Tianshuai Hu², Ronghe Qiu¹, Junwei Liang^{1,2*}

¹ AI Thrust, The Hong Kong University of Science and Technology (Guangzhou)
² Department of Computer Science and Engineering, The Hong Kong University of

Science and Technology
*Corresponding author

**All authors are my students and
I'm the only advisor.
Work done in almost 8 months.**

<https://zeying-gong.github.io/projects/falcon/>

Social Navigation - 社交导航主题研究 - ICRA 2025

Third-party Media report: 量子位报道: https://mp.weixin.qq.com/s/NT8Wk6N3S_mQwN_f39Arog

我们的机器人演示：社交导航

基于人体轨迹预测的社交导航 [3]

- 利用视觉信息预测人体轨迹[1,2]，确保机器人规划的路径遵循社交礼仪。保障 AI 机器人的身心安全
- 机器人正前往盆栽。注意机器人如何后退以避开行人



[1] Liang, Junwei, et al. "The garden of forking paths: Towards multi-future trajectory prediction." CVPR 2020.
[2] Liang, Junwei, et al. "Peeking into the future: Predicting future person activities and locations in videos." CVPR 2019
[3] "From Cognition to Precognition: A Future-Aware Framework for Social Navigation", ICRA 2025

Our robot demos: Social Navigation

Social Navigation using Human Trajectory Prediction [3]

- Predict human trajectory using visual information^[1,2], make sure robot plans a path that follows social etiquette. Ensure physical and mental safety of AI robots
- **The robot is going to the plant.** See here how the robot **moves backward** to avoid the person

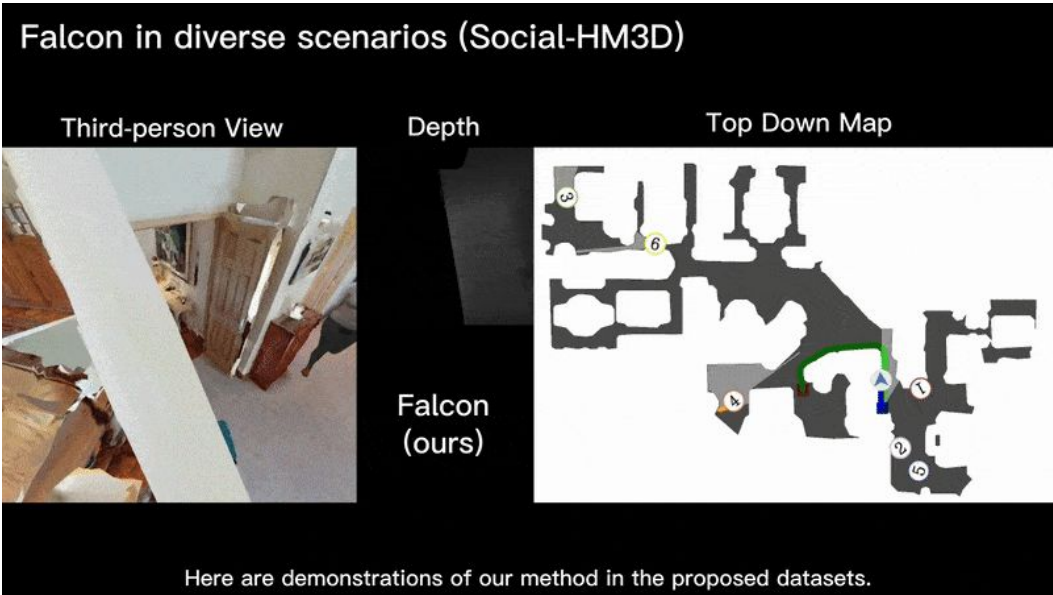


[1] Liang, Junwei, et al. "The garden of forking paths: Towards multi-future trajectory prediction." CVPR 2020.
[2] Liang, Junwei, et al. "Peeking into the future: Predicting future person activities and locations in videos." CVPR 2019
[3] "From Cognition to Precognition: A Future-Aware Framework for Social Navigation", ICRA 2025

我们的机器人演示：社交导航

基于人体轨迹预测的社交导航

○更多示例——你们将能在 Project 3 作业中亲自训练该模型！

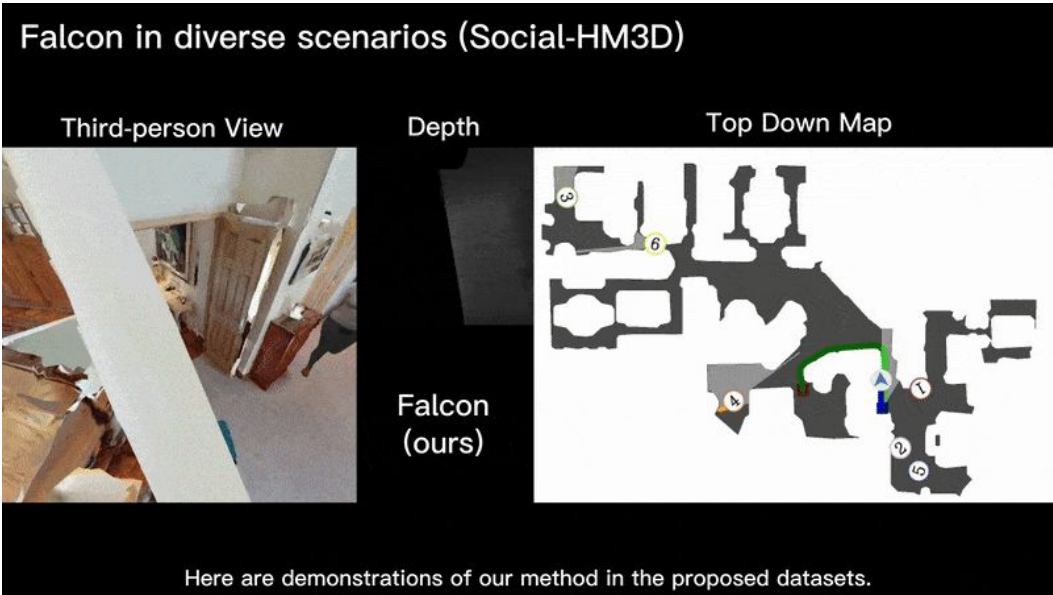


[1] Liang, Junwei, et al. "The garden of forking paths: Towards multi-future trajectory prediction." CVPR 2020.
[2] Liang, Junwei, et al. "Peeking into the future: Predicting future person activities and locations in videos." CVPR 2019
[3] "From Cognition to Precognition: A Future-Aware Framework for Social Navigation", ICRA 2025

Our robot demos: Social Navigation

Social Navigation using Human Trajectory Prediction

○ More examples - You will be able to train this model yourselves for Project 3 homework!



[1] Liang, Junwei, et al. "The garden of forking paths: Towards multi-future trajectory prediction." CVPR 2020.
[2] Liang, Junwei, et al. "Peeking into the future: Predicting future person activities and locations in videos." CVPR 2019
[3] "From Cognition to Precognition: A Future-Aware Framework for Social Navigation", ICRA 2025

人形机器人（Unitree G1）上的真实世界实验：

FLUX on a Humanoid Robot (4x Speed)



Predicted Traj.

Selected Traj.

RGB

Depth

Real-World Experiment on a Humanoid Robot (Unitree G1):

FLUX on a Humanoid Robot (4x Speed)



Predicted Traj.

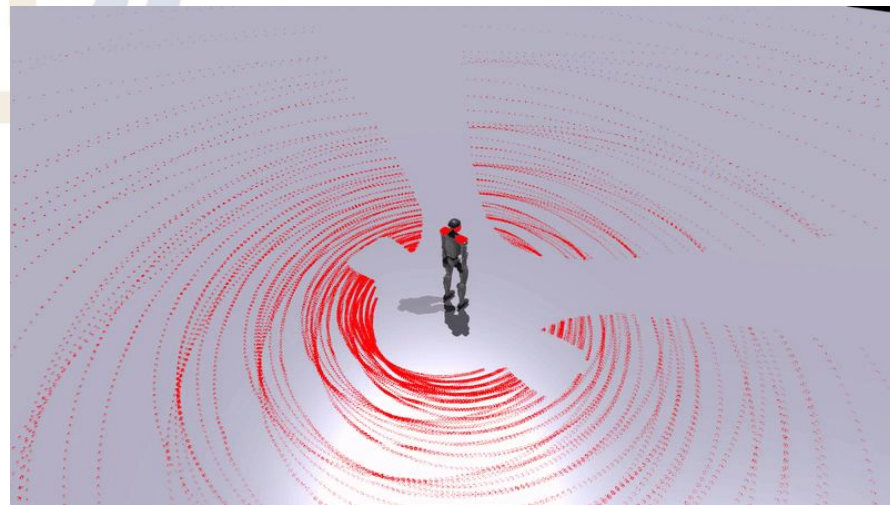
Selected Traj.

RGB

Depth

我们的机器人演示：敏捷运动

- 我们也可以利用强化学习和 3D 仿真器训练“运动策略”，使机器人避开移动障碍

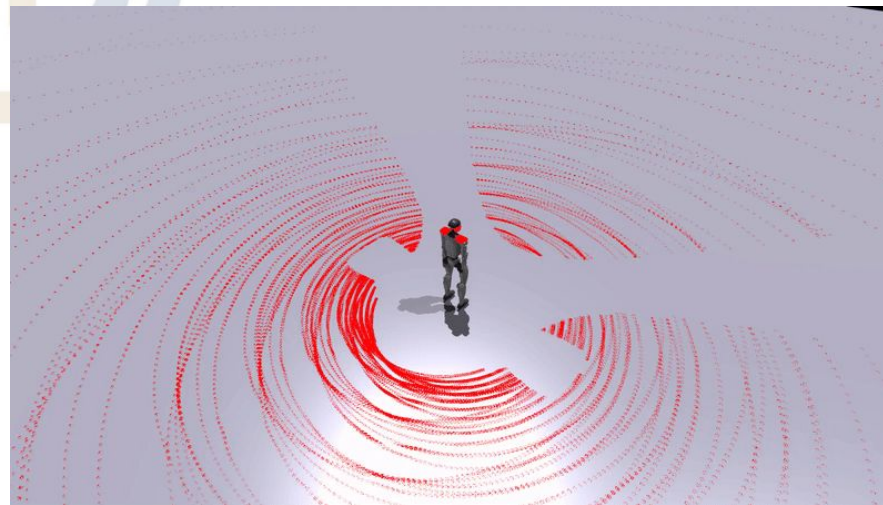


*此为 0.5 倍速回放

<https://acodedog.github.io/OmniPerceptionPages/>

Our robot demos: Agile Locomotion

- We can also utilize reinforcement learning with 3D simulator to train a “locomotion policy” that allows robot to avoid moving obstacles



*This is 0.5x playback speed

<https://acodedog.github.io/OmniPerceptionPages/>

具身智能实验室 / Precognition Lab - Overview

- 目标：构建人类水平的 AI 系统，使其能够有效感知、推理并与现实世界交互，造福人类
 - 最终得到能帮助我们完成日常任务的通用家用机器人

空间智能 ☒ VLM

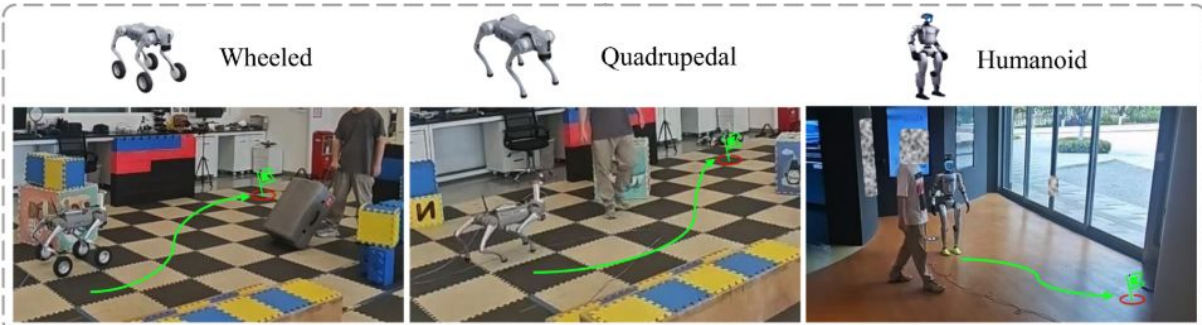
“行动”

移动能力：导航 + 运动控制

操作

“如何做事”

“如何到达某处”



实验室网站: <https://precognition.team/>

具身智能实验室 / Precognition Lab - Overview

- Goal: Build human-level AI systems that could effectively perceive, reason and interact with the real-world for the good of humans
 - So eventually we have a **general household robot** that can help us with everyday tasks

Spatial Intelligence → VLM

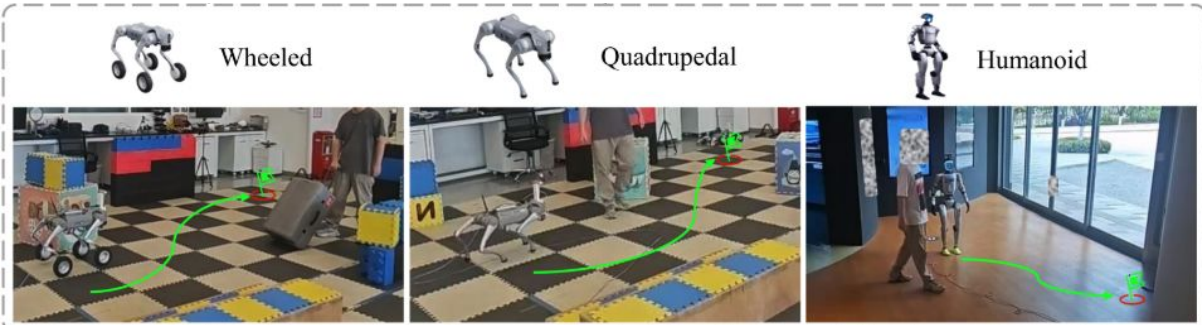
“Action”

Mobility - Navigation + Locomotion

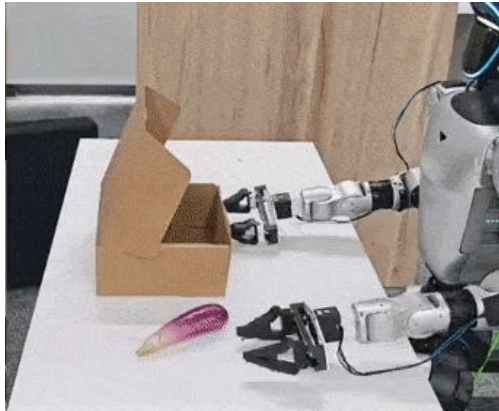
Manipulation

“How to do stuff”

“How to get somewhere”

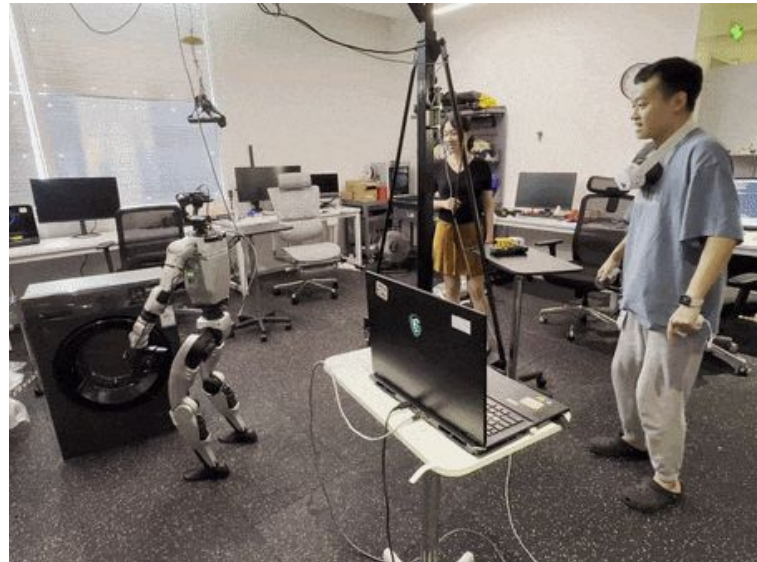


Lab website: <https://precognition.team/>



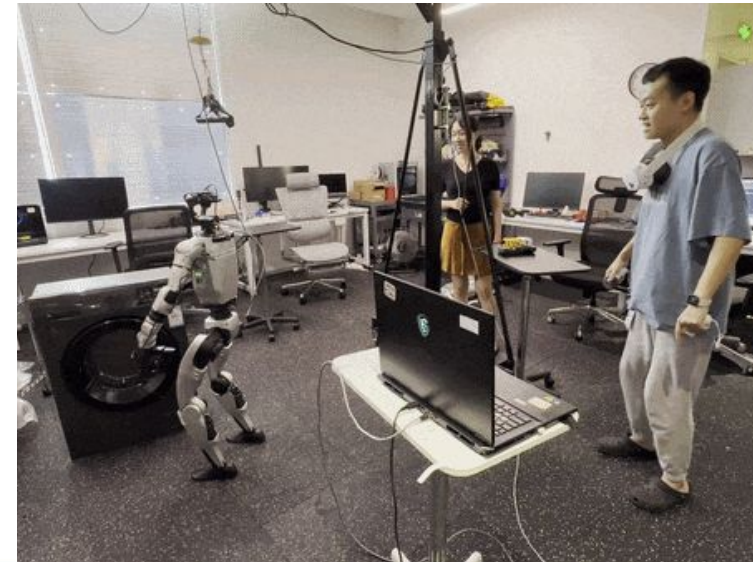
全身控制遥操作

- 基于遥操作的数据采集：获取全身控制数据
需要人类操作员通过 VR 映射关节级指令，这本质上需要一个稳健的基线基础模型
- 采集到的机器人数据与视觉数据可用于训练 VLA 模型



Whole-body-control TeleOperation

- Teleoperation-Based Data Collection: Acquiring whole-body control data requires human operators to map joint-level commands via VR, which inherently necessitates a robust baseline foundation model
- The collected robot data with vision data can be used to train VLA models



MotionWAM: Towards Foundation World Action Models for Real-Time Humanoid Loco-Manipulation 基于视频生成世界模型的人形机器人整身操作

Jia Zheng^{1,2,†} Teli Ma^{1,2,†} Yudong Fan¹ Zifan Wang^{1,2} Shuo Yang^{1,*} Junwei Liang^{2,3,*}

¹Mondo Robotics ²HKUST(GZ) ³HKUST

[†]Equal contribution. ^{*}Corresponding author, Co-advising.

<https://dit4dit.github.io/MotionWAM/>

Robotic Manipulation - 机器人操作研究 - ArXiv 2026
多家公众号争先报道, 包括具身智能研究室、具身智能之心等

MotionWAM: Towards Foundation World Action Models for Real-Time Humanoid Loco-Manipulation 基于**视频生成世界模型**的人形机器人**整身操作**

Jia Zheng^{1,2,†} Teli Ma^{1,2,†} Yudong Fan¹ Zifan Wang^{1,2} Shuo Yang^{1,*} Junwei Liang^{2,3,*}

¹Mondo Robotics ²HKUST(GZ) ³HKUST

[†]Equal contribution. ^{*}Corresponding author, Co-advising.

<https://dit4dit.github.io/MotionWAM/>

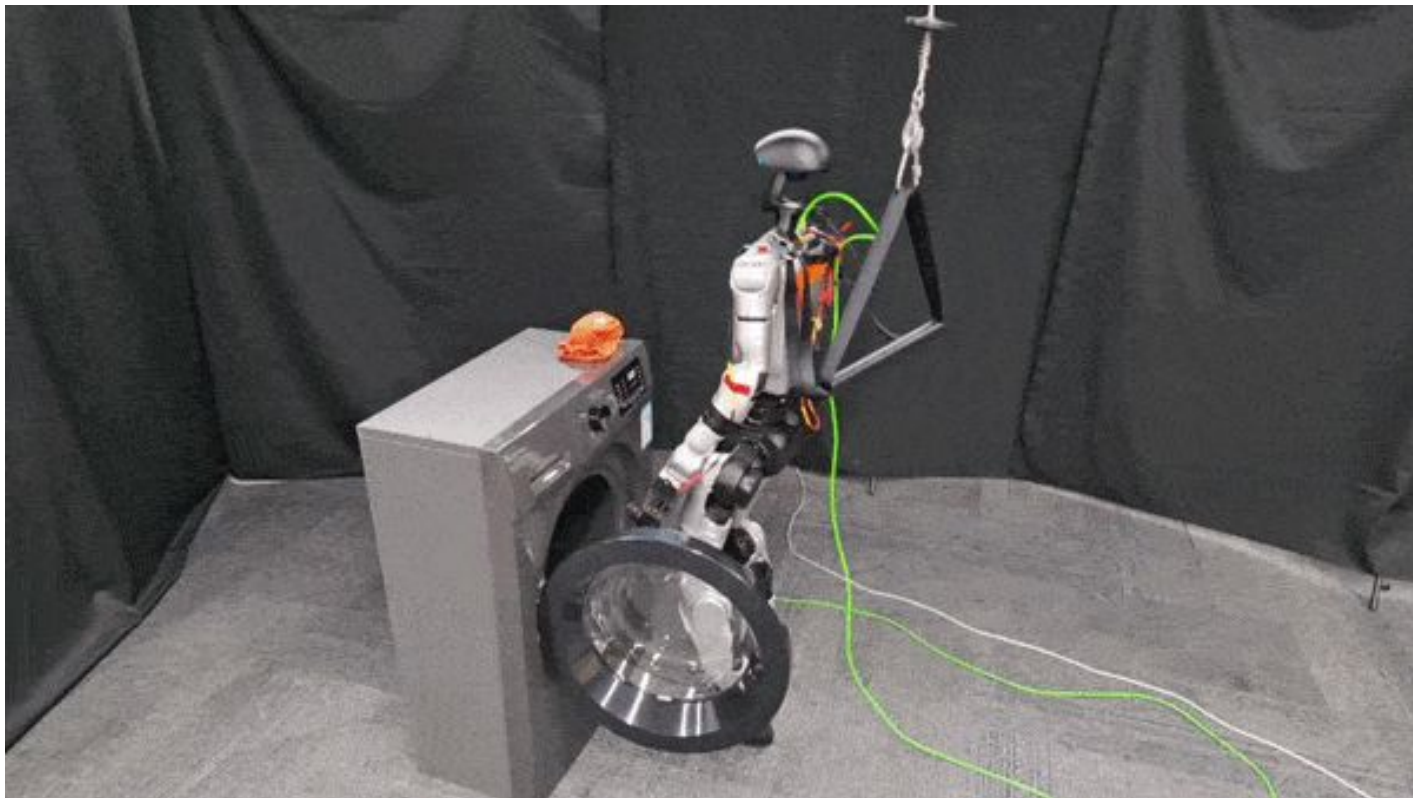
Robotic Manipulation - **机器人操作** 研究 - ArXiv 2026
多家公众号争先报道, 包括具身智能研究室、具身智能之心等

真实机器人实验——洗衣服

- 策略甚至发现门没有关好并把它关上

完全自主，
无加速，无
遥操作。

我们的
MotionWAM-2.5B
可在 A100 GPU 上
以 5Hz 运行，
解码 100 步
动作块



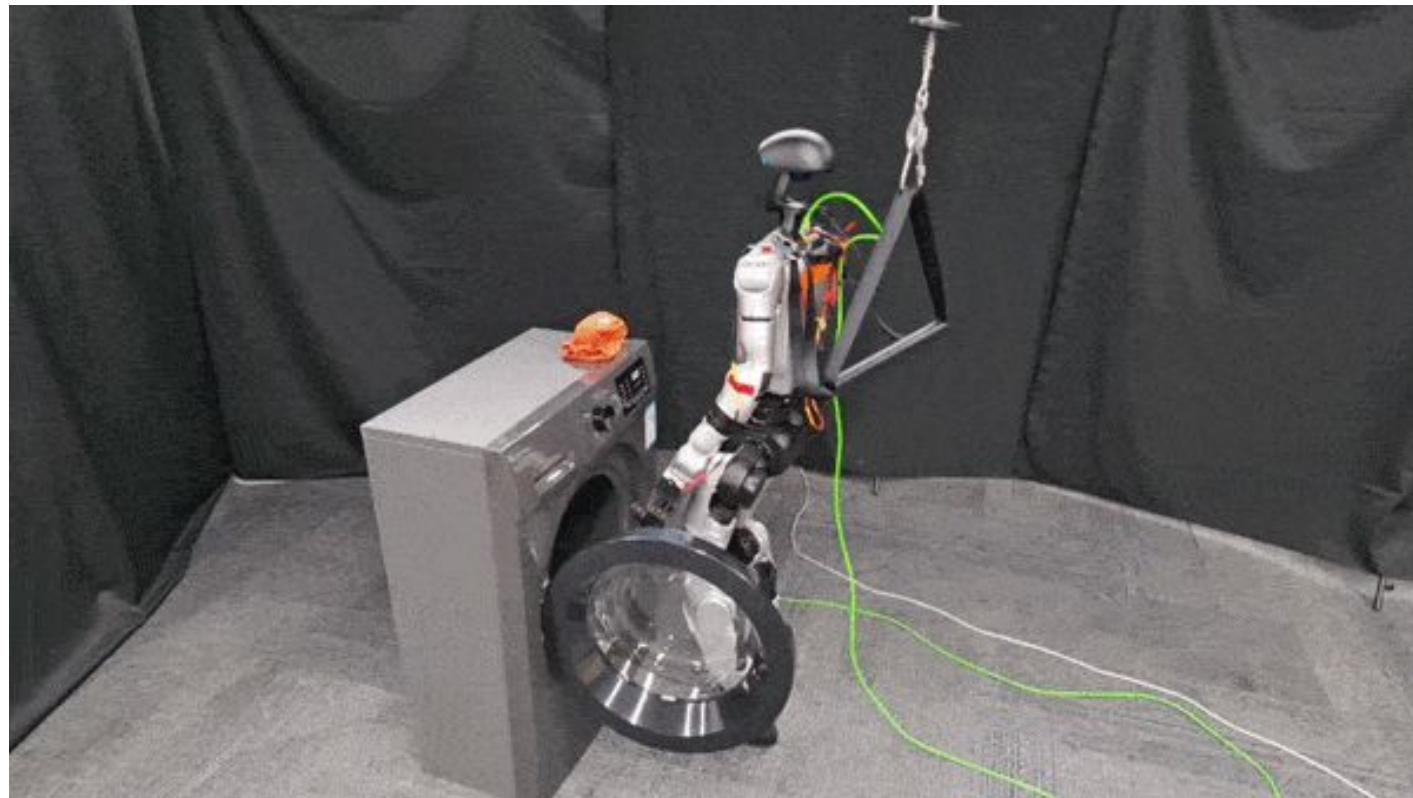
55

Real robot experiments - Do laundry

- The policy even realizes that the door has not been properly closed and fixes it

Fully autonomous,
no speed-up, no
teleoperation.

Our
MotionWAM-2.5B
can run at 5Hz on a
A100 GPU to
decode a 100-step
action chunk



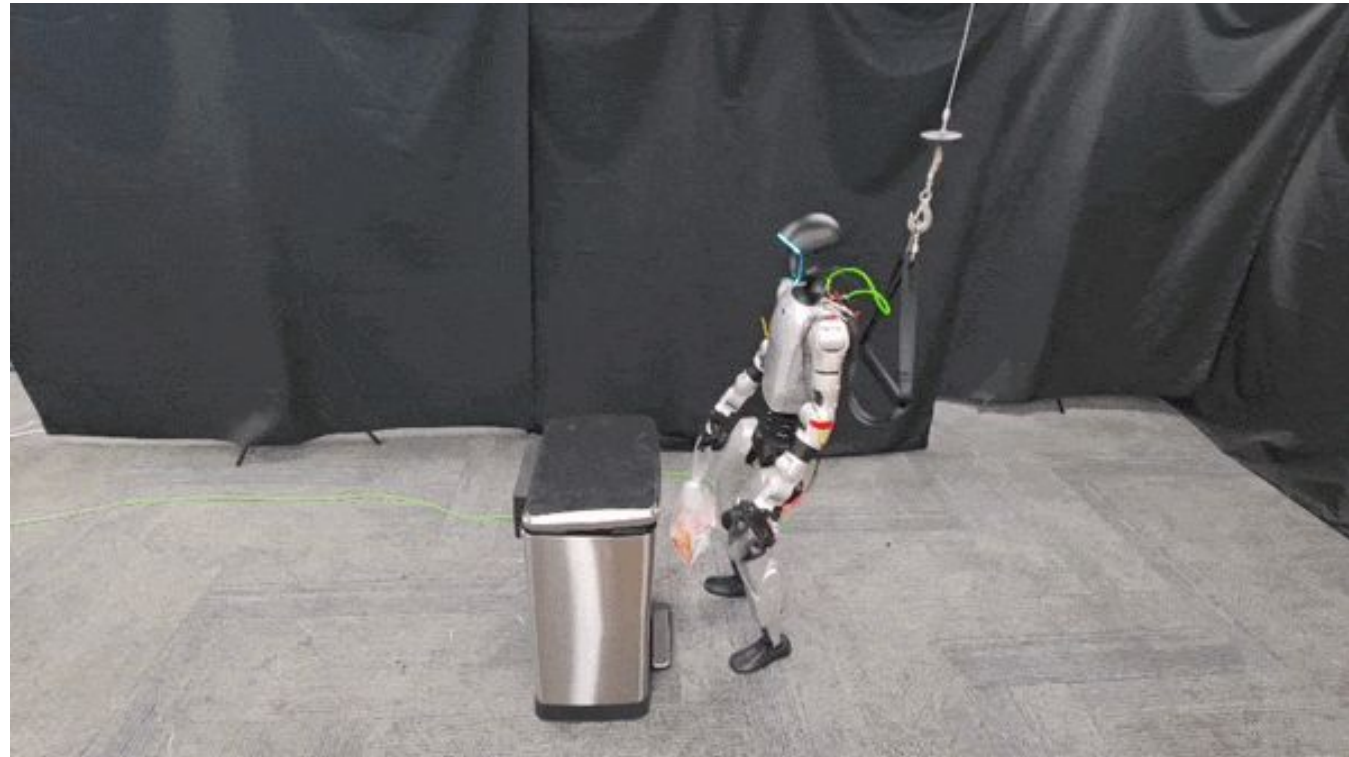
55

真实机器人实验——扔垃圾

- 机器人必须学会精确落脚

完全自主，
无加速，无
遥操作。

我们的
MotionWAM-2.5B
可在 A100 GPU 上
以 5Hz 运行，
解码 100 步
动作块



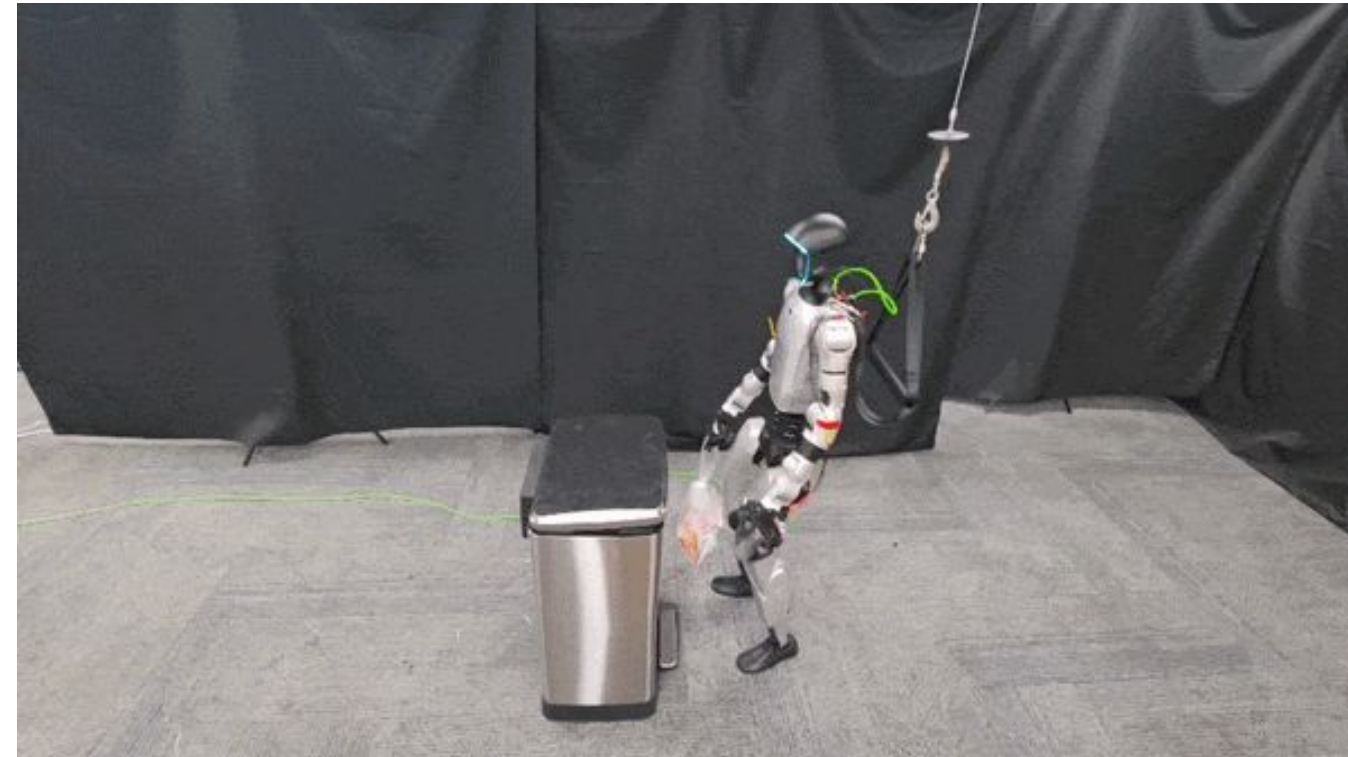
56

Real robot experiments - Throw away trash

- The robot has to learn precise foot placement

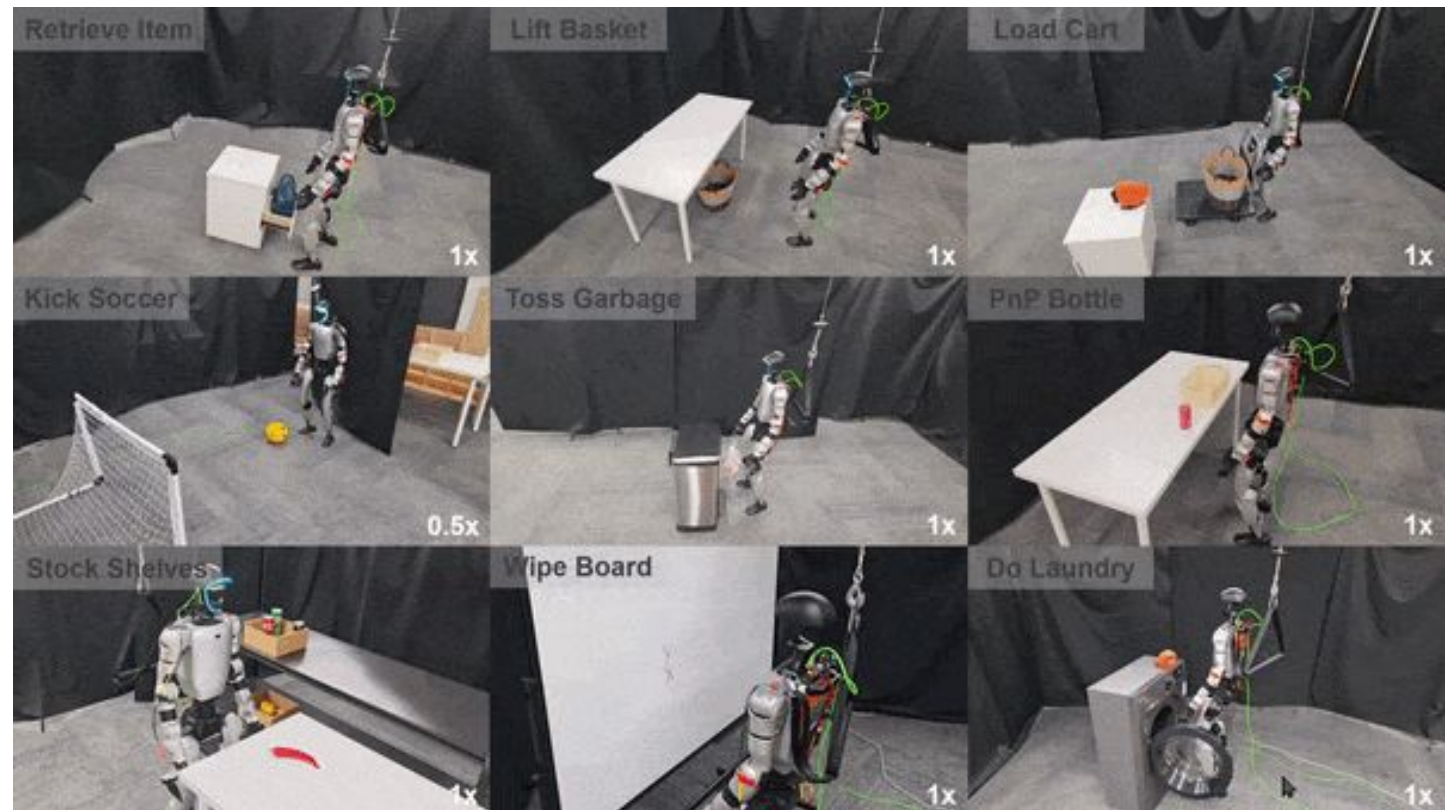
Fully autonomous,
no speed-up, no
teleoperation.

Our
MotionWAM-2.5B
can run at 5Hz on a
A100 GPU to
decode a 100-step
action chunk



56

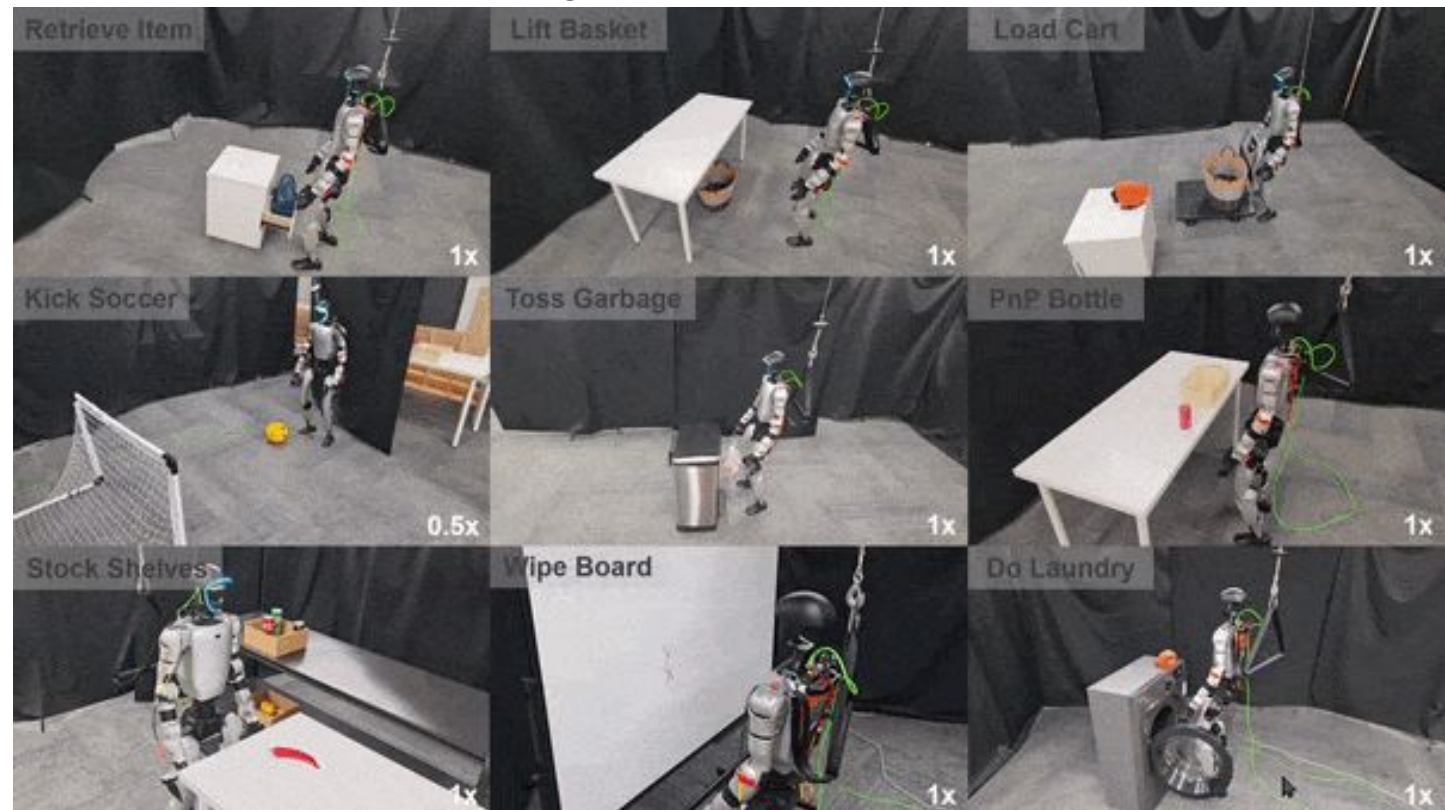
全部 9 个真实世界任务



57

All 9 Real World Tasks

- More videos at: <https://dit4dit.github.io/MotionWAM/>



57

具身智能实验室 / Precognition Lab - Overview

- 目标：构建人类水平的 AI 系统，使其能够有效感知、推理并与现实世界交互，造福人类
 - 最终得到能帮助我们完成日常任务的通用家用机器人

空间智能 ☒ VLM

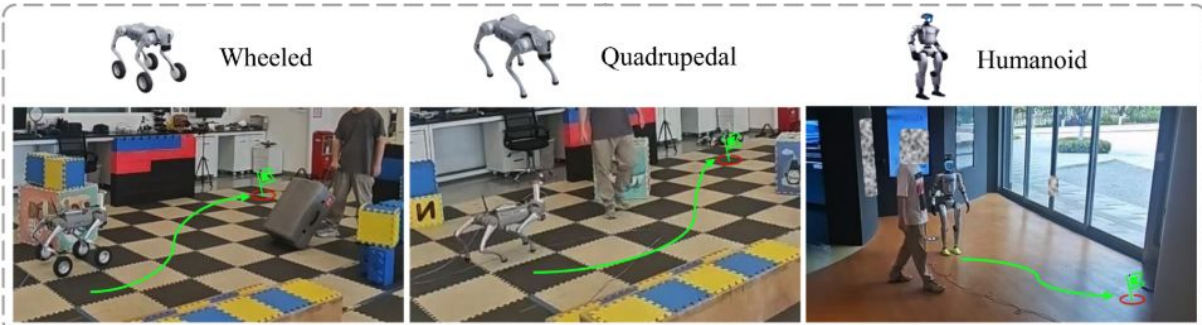
“行动”

移动能力：导航 + 运动控制

操作

“如何做事”

“如何到达某处”



实验室网站: <https://precognition.team/>

具身智能实验室 / Precognition Lab - Overview

- Goal: Build human-level AI systems that could effectively perceive, reason and interact with the real-world for the good of humans
 - So eventually we have a **general household robot** that can help us with everyday tasks

Spatial Intelligence → VLM

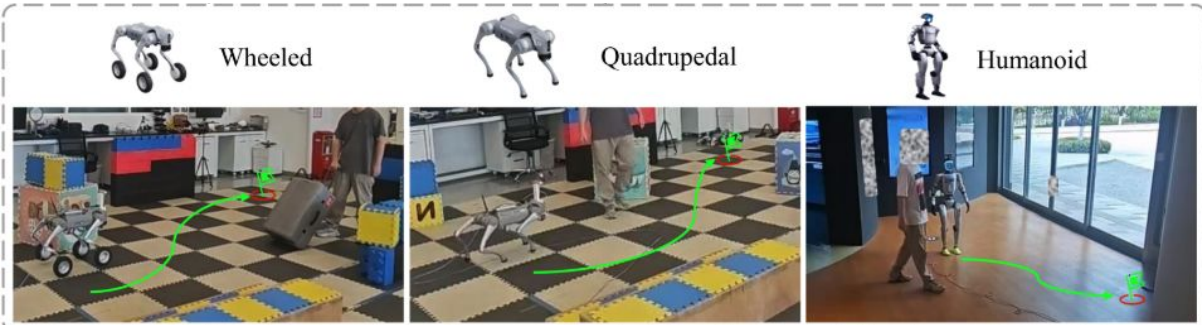
“Action”

Mobility - Navigation + Locomotion

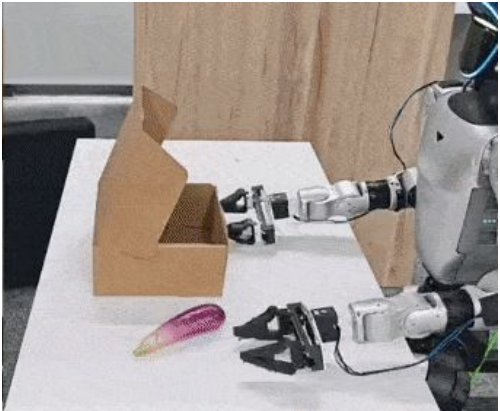
Manipulation

“How to do stuff”

“How to get somewhere”



Lab website: <https://precognition.team/>



端到端 SpeechVLA 人形智能体（已部署于大学访客中心）

- 我们的高考演示视频在 Bilibili 获得 15 万+ 播放
 - <https://www.bilibili.com/video/BV1U67V6PEGc/>



*查看完整
无剪辑演示：
这里

End-to-end SpeechVLA Humanoid Agent (Deployed at Uni. Visitor Center)

- Our demo video for Gaokao has received 150k+ views on bilibili
 - <https://www.bilibili.com/video/BV1U67V6PEGc/>



*check out the full
demo without cuts:
[here](https://www.bilibili.com/video/BV1U67V6PEGc/)

目录

- 具身智能范式
- 具身智能系统设计
- 传感器概览
- 机器人视觉 vs. 计算机视觉
- 实验室机器人演示
- 下周——机器感知！

Agenda

- The Embodied AI paradigm
- Embodied AI System Design
- Sensors Overview
- Robot Vision vs. Computer Vision
- Robot Demos from my lab
- Next week - Machine Perception!

阅读材料

- Urmson, Chris, et al. "Tartan racing: A multi-modal approach to the darpa urban challenge." (2007).
- ROS 教程: <https://www.ros.org/blog/getting-started/>

Reading Material

- Urmson, Chris, et al. "Tartan racing: A multi-modal approach to the darpa urban challenge." (2007).
- ROS tutorials: <https://www.ros.org/blog/getting-started/>



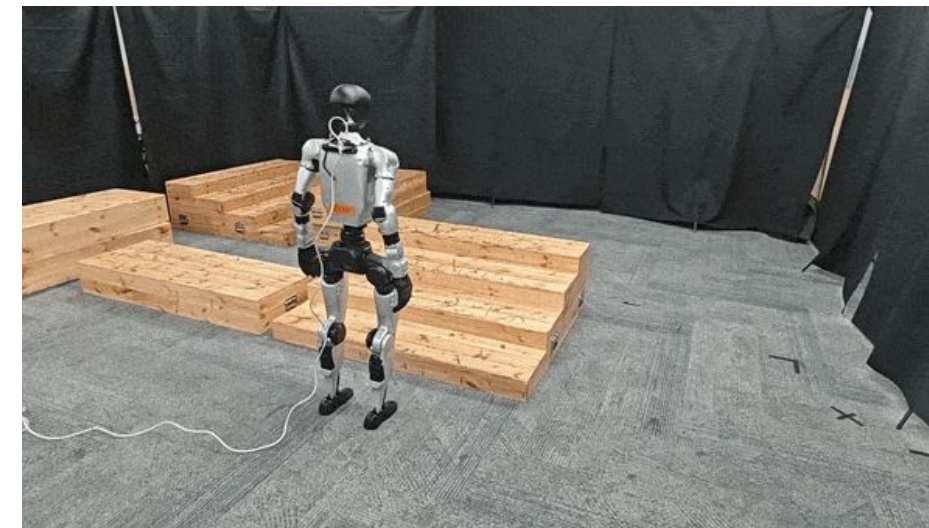
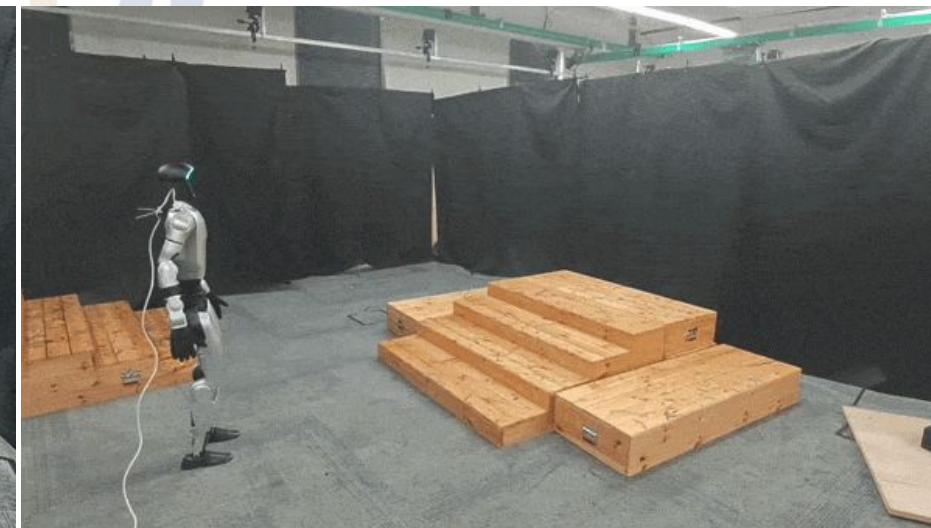
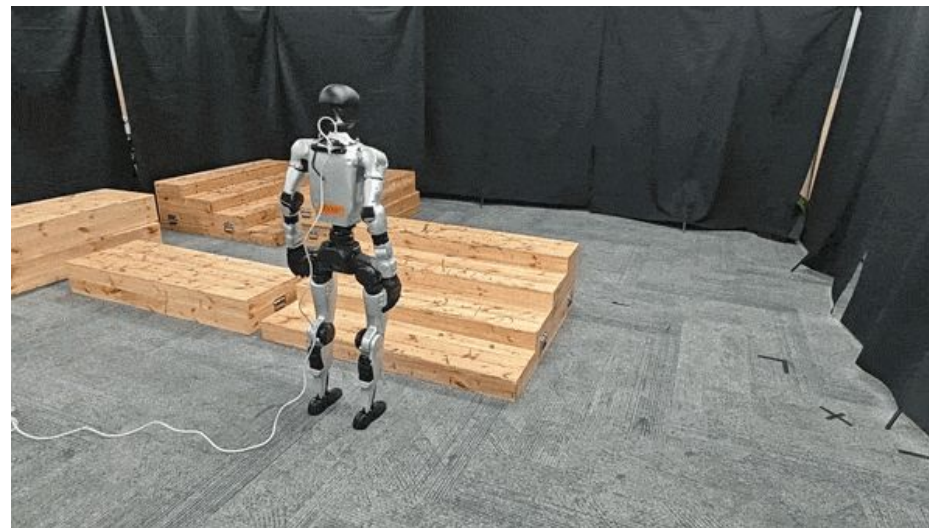
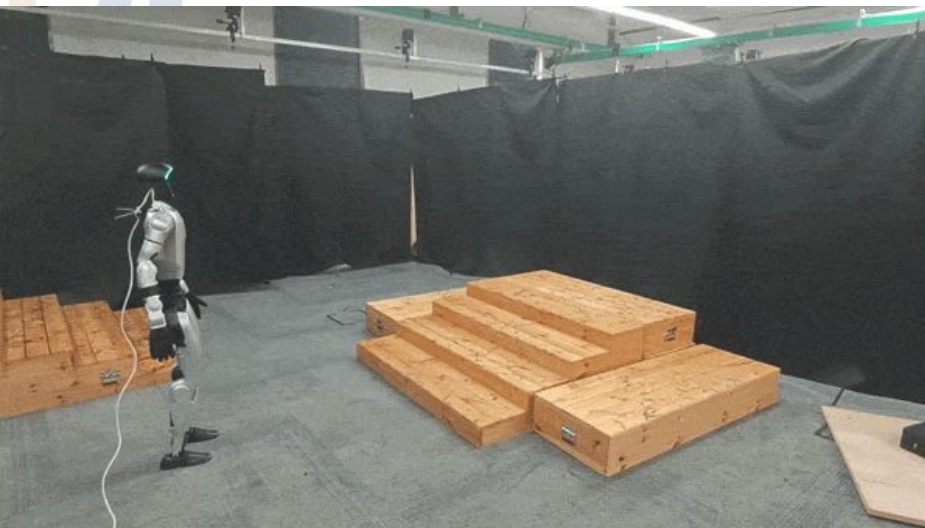
谢谢！



Thank you!

感知行为基础模型

- 我们甚至能在上楼梯时做后空翻!
 - 原始关节指令是针对平地的。



- 更多视频: <https://acodedog.github.io/perceptive-bfm/>

Perceptive Behavior Foundation Model

- We can even do backflips while walking up stairs!
 - Original joint commands are for flat grounds.

- More videos: <https://acodedog.github.io/perceptive-bfm/>